

treatments, up 100% from eight cities as of the 2010 Benchmarking Report. Eight cities report implementing home zones, or woonerfs (only San Jose previously reported experimenting with home zones).

New to the 2012 Benchmarking Report, cities were also asked about additional innovative treatments including bike boxes, cycle tracks, and contra flow bike lanes. Twenty cities report having installed bike boxes, or advanced stop lines, which prioritize cyclists at red lights. Eleven cities have installed cycle tracks and ten cities have contra flow bike lanes.

Portland has used more innovative treatments than any other major U.S. city having implemented every innovation surveyed. San Francisco is close behind only lacking bicycle boulevard implementation. Long Beach, Minneapolis, New York, and Seattle also lead for innovative facilities with six of eight innovative facilities surveyed.

U.S. Bicycle Route System

Also new to the 2012 Benchmarking Report are data about state involvement with the U.S. Bicycle Route System. The U.S. Bicycle Route System (USBRS) is a proposed national network of bicycle routes. These routes link urban, suburban, and rural areas with appropriate bicycle friendly routes including trails, bike paths, roads with shoulders, and low-traffic routes. For a route to be designated as part of the USBRS it must either connect two or more states, a state and an international border, or one or more U.S. Bicycle Routes.

The first two U.S. Bicycle Routes were designated in 1982 and then no ad-

ditional routes were nominated. In 2003 the American Association of State Highway and Transportation Officials (AASHTO) revived the USBRS with an official task force. An inventory of existing bicycle routes (see map on page 106) throughout the United States was created as a first step in drafting a national bicycle network plan. In 2008, AASHTO passed a resolution in support of the National Corridor Plan. An application for route designation was completed in May 2009 (Adventure Cycling Association, 2009).

According to data from Adventure Cycling Association, 34 states have an active USBRS program. Thirty-one states have identified potential USBRs in state or local bicycle plans. Routes have been officially designated as part of the USBRS in nine states, and three states have posted and signed USBRs.

Bike-Transit Integration

The last bicycling provision measured was bike-transit integration. This report sought to measure how well cities provide for bicyclists on transit. While most cities are successfully integrating bicycles with buses, many fall behind in regard to providing parking for bicyclists at transit (Pucher and Buehler 2009). Almost all cities surveyed have 100% of their city bus fleet equipped with bicycle racks. When it comes to bicycle parking, cities report that only 27% of transit stops have bike parking on average. According to the American Public Transportation Association, this amounts to just 2.5 bicycle parking spaces at bus stops per 10,000 residents (APTA 2011).

Innovative Facilities in Cities

City	Shared lane markings	Bicycle boulevards	Home zones/woonerfs	Colored bike lanes	Bike boxes	Cycle tracks	Contra flow bike lane	Bicycle traffic light
Albuquerque	✓★	✓★			✓			
Arlington, TX								
Atlanta	✓★							
Austin	✓★	(3)		✓★	✓	✓	(3)	(3)
Baltimore	✓	(2)		✓★	✓		✓	
Boston	✓			✓★	✓	✓		
Charlotte	✓★							
Chicago	✓		✓★	✓			✓	
Cleveland (1)								
Colorado Springs	✓★	✓★	✓★					
Columbus	✓★	✓		✓★	✓			
Dallas								
Denver	✓	(3)			✓	✓		✓★
Detroit (1)								
El Paso					✓			
Fort Worth	✓							
Fresno								
Honolulu	✓★				✓			
Houston	✓						✓	
Indianapolis (1)								
Jacksonville	✓★							
Kansas City, MO	✓★			✓★	(2)			
Las Vegas		✓★						
Long Beach	✓	✓		✓	✓	✓		✓
Los Angeles	✓★							
Louisville	✓		✓★					
Memphis								
Mesa								✓★
Miami	✓★							
Milwaukee								
Minneapolis	✓	✓★		✓	✓	✓	✓	
Nashville	✓							
New Orleans	✓							
New York	✓			✓	✓	✓	✓	✓
Oakland	✓	✓★				✓		
Oklahoma City	✓			✓★	✓			
Omaha	✓★				✓			
Philadelphia	✓★		✓★	✓	✓		✓	
Phoenix	✓★	(2)			✓			
Portland, OR	✓	✓	✓★	✓	✓	✓	✓	✓
Raleigh	✓★							
Sacramento	✓★							
San Antonio	✓★	(2)	(2)	✓★		✓		
San Diego								
San Francisco	✓		✓★	✓★	✓	✓	✓	✓
San Jose	✓	(2)	✓	(2)		(2)		
Seattle	✓	✓	✓★	✓★	✓		✓	✓★
Tucson	✓★	✓		✓	✓			✓
Tulsa								
Virginia Beach								
Washington, DC	✓				✓	✓	✓	✓★
# of cities responding yes	37	10	8	16	20	11	10	9
Mean/Average	Yes	No	No	No	No	No	No	No

Legend:

- ✓ = Yes/has implemented innovative facility
- ★ = New since 2010 Benchmarking Report

Source: City surveys 2010/2011 **Notes:** Responses of "unknown" were taken to mean "no." (1) Unanswered survey. (2) Planned or proposed. (3) Under development at time of survey.

Innovative Facilities Defined



Shared lane markings—Often called “sharrows,” these are markings which resemble a bicycle and an arrow painted on a roadway to indicate the direction of travel for bicycles as well as motorized vehicles.



Bike box—A pavement marking that utilizes two stop lines: an advanced stop line for motor vehicles, and a stop line closer to the intersection for bicyclists. This allows bicyclists to get a headstart when the light turns green to more safely proceed ahead or make a left turn.



Bicycle boulevards—A shared roadway which is intended to give priority to bicyclists by optimizing it for bicycle traffic (through traffic calming) and discouraging some motor vehicle traffic. Many of these routes have no bike lanes so bicyclists are allowed use of full lane.



Cycle track—An exclusive bicycle facility that combines the user experience of a separated path with the on-street infrastructure of a conventional bike lane.



Home zones (woonerf zones)—These streets are designated as “shared streets” and do not prioritize the needs of motor vehicles. Rather, it is a space where pedestrians and bicyclists are the priority and motor vehicles are kept at low speeds.



Contra flow bike lane—A designated bicycle lane marked to allow bicyclists to travel against the flow of traffic on a one-way street.



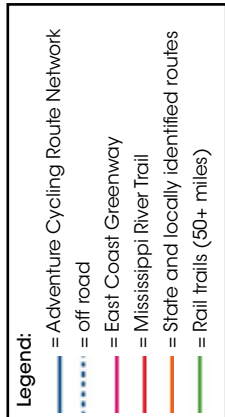
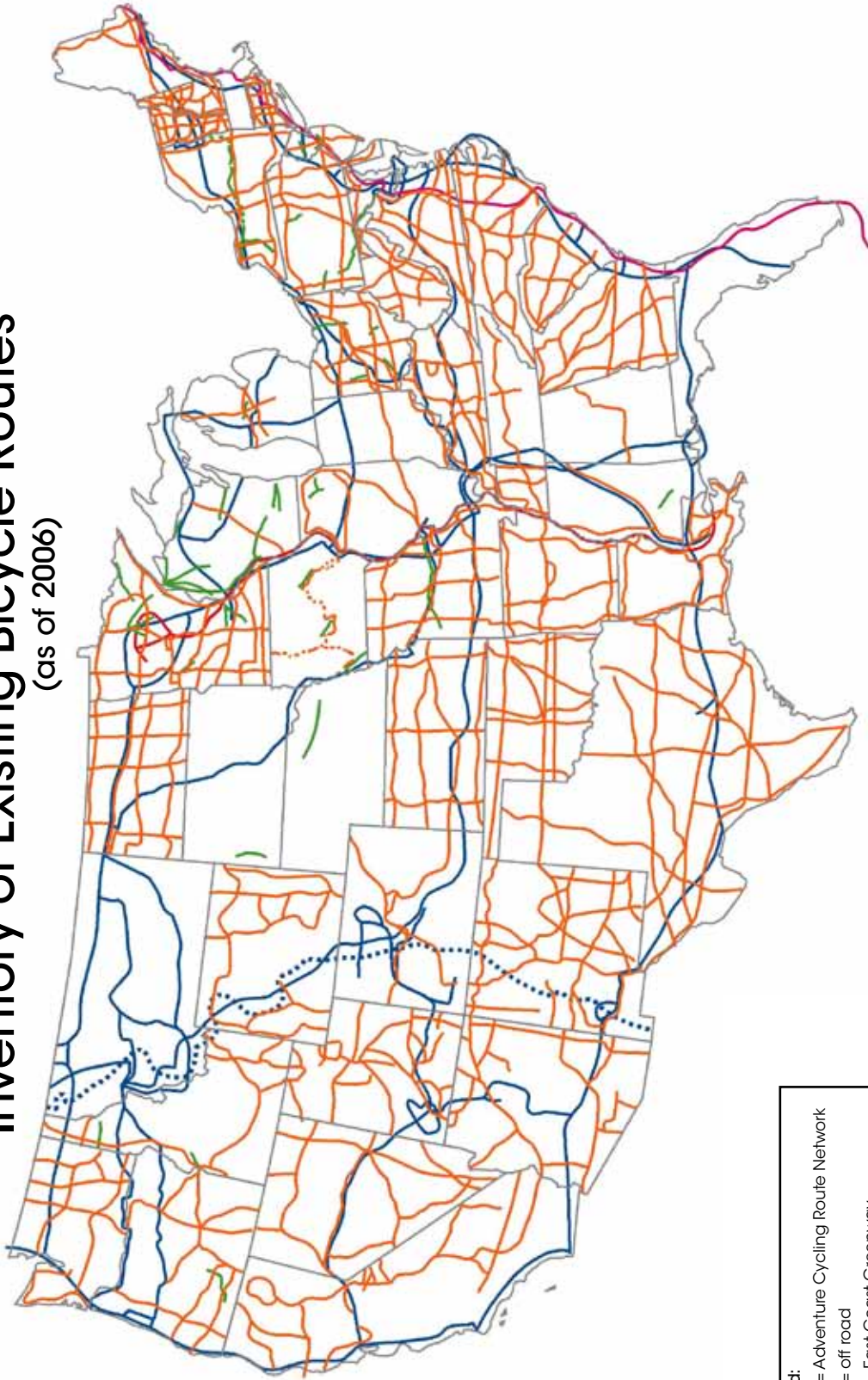
Colored bike lanes—Bike lanes which have special coloring to provide a distinct visual definition that the space is designated for bicyclists.



Bicycle traffic light—Lights on roadways which have specific bicycle symbols that illuminate to direct bicycle traffic.

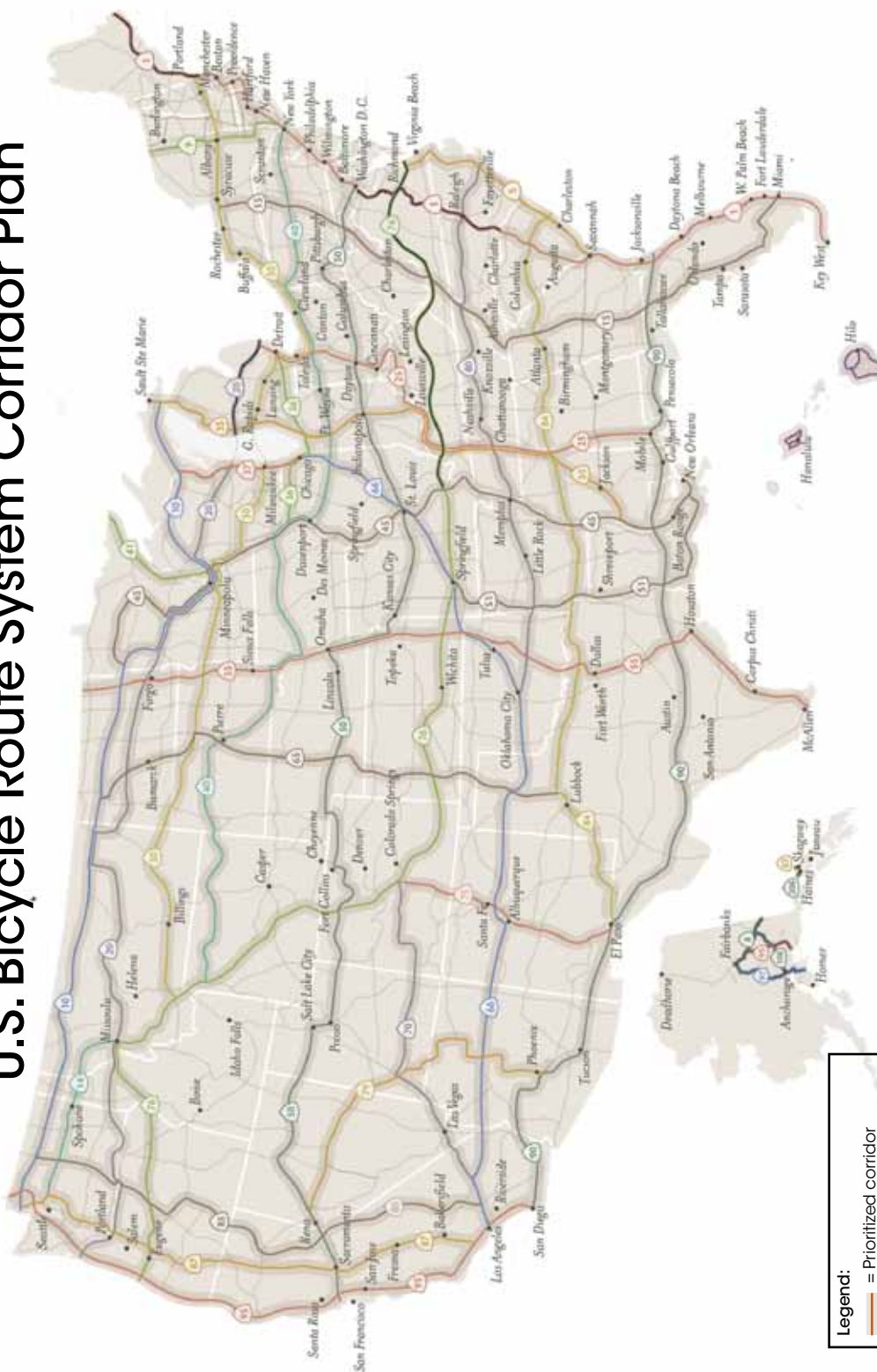
Photos top to bottom: (left) John Luton, Payton Chung, La-Citta-Vita@Flickr, Tanya Dueri—San Francisco Bicycle Coalition (right) Alrthur Wendall, New York City DOT, John Luton, Roland Tanglao

Inventory of Existing Bicycle Routes (as of 2006)



Source: Adventure Cycling Association **Note:** The inventory of existing bicycle routes, developed in 2006, served as the base layer for creating the National Corridor Map, a system-level plan for states to use when planning and designating interstate bicycle routes. This map is current as of 2006.

U.S. Bicycle Route System Corridor Plan



Source: Map courtesy of Adventure Cycling Association, September 2011 **Notes:** Prioritized corridors are not routes, but 50-mile-wide areas where a route may be developed. These corridors have been assigned route numbers. Alternate corridors provide additional consideration for interstate routing. These corridors have not been assigned route numbers but may be added or existing corridors shifted as needed. Established U.S. Bicycle Routes designated by AASHTO appear as defined lines on the Corridor Map. For specific route information visit www.adventurecycling.org/routes/usbrs.

U.S. Bicycle Route Policy

State	Active USBR program	USBRS recognized in state or local bike plan	One or more USBR(s) in state received official AASHTO designation	Has one or more signed USBRs
Alabama		✓		
Alaska	✓		✓	
Arizona	✓			
Arkansas		✓		
California				
Colorado				
Connecticut	✓	✓		
Delaware	✓	✓		
Florida	✓	✓	✓	
Georgia	✓	✓		
Hawaii				
Idaho				
Illinois	✓	✓	✓	
Indiana		✓		
Iowa	✓	✓		
Kansas				
Kentucky	✓	✓	✓	✓
Louisiana	✓	✓		
Maine	✓	✓	✓	
Maryland	✓	✓		
Massachusetts	✓	✓		
Michigan	✓	✓	✓	
Minnesota	✓	✓		
Mississippi	✓			
Missouri	✓	✓		
Montana				
Nebraska				
Nevada	✓	✓		
New Hampshire	✓		✓	
New Jersey				
New Mexico	✓	✓		
New York	✓	✓		
North Carolina	✓	✓	✓	✓
North Dakota	✓	✓		
Ohio	✓	✓		
Oklahoma				
Oregon	✓			
Pennsylvania	✓	✓		
Rhode Island	✓	✓		
South Carolina		✓		
South Dakota	✓	✓		
Tennessee	✓	✓		
Texas				
Utah		✓		
Vermont				
Virginia	✓	✓	✓	✓
Washington	✓			
West Virginia	✓			
Wisconsin	✓	✓		
Wyoming	✓			
# of states responding yes	34	31	9	3
Mean/Average	Yes	Yes	No	No

Sources: Adventure Cycling Association October 2011
 Photo: A rare duplex of US Bike Routes 1 and 76, in downtown Ashland, VA.
 Photo by Will Weaver.

Legend:

✓ = Yes/has policy



Bike-Transit Integration



Photo by Metro Cincinnati Metro Bus

Most bus fleets in major cities are equipped for bicycles.

Forty-three cities report that 100% of their city buses are equipped with bicycle racks. This is up 43% from 2007 when just 30 cities reported that 100% of their buses had racks. New York remains the only major U.S. city with no bicycle racks on buses. Regarding bicycle parking at bus stops, cities average 1.3 bicycle parking spaces for every 10,000 residents. This is up 8% since the 2010 Benchmarking Report.

Legend:

∅ = Not applicable
 * = Officials could not access data
 ■ = High value ■ = Low value

City	% of buses with bicycle racks	Hrs bikes allowed on trains/ hrs trains operated	Bike parking at transit		
			# of bicycle parking spaces at bus stops(1)	Bike parking spaces per 10,000 people(1)	% of transit stops with bicycle parking
Albuquerque	100%	105/105	*	*	100.0%
Arlington, TX	∅	∅	*	*	∅
Atlanta	100%	139.25/139.25	*	*	*
Austin	100%	40/40	156	1.97	1.0%
Baltimore	100%	168/168	*	*	*
Boston	50%	98/140	0	0.00	*
Charlotte	100%	137/137	51	0.72	*
Chicago	100%	148/168	*	*	92.0%
Cleveland	*	*	6	0.14	*
Colorado Springs	100%	∅	0	0.00	*
Columbus	100%	∅	54	0.70	0.6%
Dallas	100%	140/140	15	0.12	*
Denver	99%	168/168	682	11.17	15.0%
Detroit	25%	*	*	*	0.0%
El Paso	100%	*	74	1.19	*
Fort Worth	100%	107.5/107.5	18	0.25	0.2%
Fresno	100%	∅	30	0.63	5.0%
Honolulu	100%	∅	18	0.48	0.5%
Houston	95%	*	442	1.95	90.0%
Indianapolis	100%	*	*	*	*
Jacksonville	100%	∅	131	1.61	12.0%
Kansas City, MO	98%	∅	18	0.37	0.0%
Las Vegas	100%	∅	*	*	*
Long Beach	100%	140/*	28	0.61	*
Los Angeles	100%	130/150	272	0.71	83.5%
Louisville	100%	∅	0	0.00	*
Memphis	100%	107/107	0	0.00	0.1%
Mesa	100%	100/100	0	0.00	60.0%
Miami	100%	113/113	0	0.00	*
Milwaukee	100%	0/60	*	*	*
Minneapolis	100%	168/168	301	7.81	*
Nashville	100%	34/34	0	0.00	25.0%
New Orleans	100%	∅	*	*	1.0%
New York	0%	168/168	8	0.01	8.9%
Oakland	100%	120/140	0	0.00	92.0%
Oklahoma City	100%	∅	0	0.00	*
Omaha	100%	∅	*	*	0.0%
Philadelphia	100%	103/133	*	*	*
Phoenix	100%	140/140	54	0.34	*
Portland, OR	100%	140/140	420	7.41	6.0%
Raleigh	100%	∅	*	*	0.5%
Sacramento	100%	*	*	*	*
San Antonio	100%	∅	8	0.06	0.8%
San Diego	100%	152/152	0	0.00	*
San Francisco	100%	100/120	544	6.67	*
San Jose	100%	145/145	259	2.68	*
Seattle	100%	131/131	2,390	38.76	*
Tucson	100%	∅	59	1.08	*
Tulsa	100%	∅	*	*	1.0%
Virginia Beach	100%	∅	*	*	1.0%
Washington, DC	100%	105/135	*	*	100.0%(3)
Mean/Average	95%	91%	173	2.5 (2)	26.8%
Median	100%	100%	18	0.3	3.0%
High	100%	100%	2,390	61.4	100.0%
Low	0%	0%	0	0.0	0.0%

Sources: City surveys 2010/2011 (1) APTA 2011 Note: (2) Average weighted. (3) 100% of rail stops have bike parking; percentage for bus stops unknown.



Children line up for a kids' bicycle race. Photo courtesy of Paul Dineen

5: EDUCATION AND ENCOURAGEMENT

2 of the "5 Es"

Education and encouragement are two of the often cited "5 Es" needed for making a community bicycle and pedestrian friendly. Both bicyclists and motorists need education on how to safely share the road and navigate traffic. Wide-spread education efforts can contribute to safer roadways for all. Encouragement is also needed to promote the spread of bicycling and walking as means of transport, recreation, and physical activity.

The 2010 Benchmarking Report was the first to establish benchmarks for bicycle and pedestrian education and encouragement efforts. Many states and cities

have implemented programs and events with these aims but have had no way to evaluate their success compared to others. This report builds on data from the previous report to track progress of these efforts.

The "5 Es"

1. Engineering
2. Education
3. Encouragement
4. Enforcement
5. Evaluation

Educating Professionals

This report measured education efforts in two areas: professional education and public education. The first refers to the education professionals receive that contributes to the promotion and safety of bicycling and walking. Included is

Bike/Ped Professional Education in States

State	Bicycling enforcement police academy requirement	Bicycling enforcement police continuing training	Annual statewide bike/ped conference (1)
Alabama		✓	
Alaska			
Arizona		✓	
Arkansas			
California	✓	✓	✓ (3)
Colorado			✓ ★ (2,4)
Connecticut		✓	
Delaware			✓ (2)
Florida		✓	✓ (5)
Georgia		✓	✓ ★ (2)
Hawaii			
Idaho			
Illinois	✓	✓	
Indiana	✓	✓	✓ (2)
Iowa	✓	✓	✓ (2)
Kansas		✓	✓ ★
Kentucky	✓	✓	✓ ★
Louisiana	✓	✓	
Maine		✓	✓
Maryland		✓	✓
Massachusetts	✓	✓	✓
Michigan		✓	✓ (2)
Minnesota		✓	✓ (2)
Mississippi	✓		
Missouri	✓	✓	(7)
Montana			
Nebraska		✓	
Nevada		✓	✓
New Hampshire	✓		
New Jersey	✓		✓ ★
New Mexico		✓	
New York		✓	
North Carolina		✓	
North Dakota			
Ohio	✓	✓	✓ ★
Oklahoma		✓	
Oregon	✓	✓	✓ ★
Pennsylvania		✓	✓ ★
Rhode Island	✓	✓	✓ ★
South Carolina	✓	✓	(7)
South Dakota			✓ ★ (2)
Tennessee		✓	
Texas	✓	✓	✓ ★ (3)
Utah		✓	✓ ★ (2)
Vermont	✓		
Virginia		✓	
Washington	✓	✓	✓ ★ (3,6)
West Virginia		✓	(7)
Wisconsin	✓	✓	✓ (3)
Wyoming	✓		
# of states responding yes	20	36	25
Mean/Average	No	Yes	ø

Professional education on bicycling and walking is growing.

Twenty states (40%) report that bicycling enforcement is a police academy requirement. This is up from just 11 states as of the 2010 Benchmarking Report. Twenty-five states report having a statewide bicycle and pedestrian conference (up from 16).

Sources: State surveys 2010/2011, LAB 2011

Notes: (1) Statewide bike/ped conference refers to an educational and networking event that brings together professionals working on bicycle and pedestrian issues including government officials, planners, educators, and advocates. (2) Bicycle conference only. (3) Biennial. (4) Hosted by Bicycle Colorado. (5) Annual conference 2005-2010; is not planned for 2011 because of budget cuts; may be reconsidered in future. (6) Trails conference includes bicycle and pedestrian issues. (7) Former conference no longer active.

Legend:

✓ = Yes

ø = Not applicable

★ = New since 2010 Benchmarking Report

the education of police officers in bicycle laws and safety and the education of government employees and other professionals working to promote, plan, and implement policies and provisions for bicycling and walking.

Police Training

Police officers without training in bicycle laws may not understand or uphold bicyclists' or pedestrians' rights in traffic crashes, incorrectly stop or ticket bicyclists, or set a bad example of the law for other motorists. Education of law enforcement in bicycle safety and laws pertaining to bicycling is critical to furthering bicycling safety and rights.

Data on police officer education come from the League of American Bicyclists' Bicycle Friendly State surveys. According to these surveys, 20 include bicycling enforcement as a Police Academy requirement and 36 states include bicycling enforcement in their police continuing education training.

Bicycle and Pedestrian Conferences

Bicycle and pedestrian professionals need opportunities for continuing edu-

cation, networking, and collaboration to further their work and profession. Many states now hold annual bicycle and pedestrian conferences or summits that provide bicycle and pedestrian professionals an opportunity for learning, networking, and planning. Sixteen states report having hosted a statewide bicycle and pedestrian conference and another nine have hosted a statewide bicycle conference. Of these, 21 are annual and four are biennial. Colorado and Texas noted that their conferences are coordinated by statewide advocacy organizations.

Educating the Public

Educating the public is a critical component of creating bicycling and walking friendly communities. From street-side messages of share-the-road campaigns to driver's test questions, states and cities are working to promote the safety of the most vulnerable road users. For this section we relied on data from state surveys, the League of American Bicyclists' Bicycle Friendly States Program, and the National Center for Safe Routes to School. State benchmarks include whether states have a public safety (or

(Continued page 115)

A Kansas City bus wrapped with a "Bike there, walk there" advertisement.
Photo courtesy of Mid-America Regional Council, Kansas City, MO



Public Education and Events in States

State	Share the Road/ public safety campaign	Info on bicycling in driver's manual	Driver's test questions on bicycling	State-sponsored ride to promote bicycling/activity	# of schools participating in Walk to School Day
Alabama		✓			23
Alaska	✓	✓	✓		9
Arizona	✓	✓	✓		116
Arkansas					9
California		✓		✓	429
Colorado	✓	✓	✓		215
Connecticut	✓	✓	✓		17
Delaware	✓	✓		✓	10
Florida	✓	✓	✓	✓	225
Georgia	✓	✓			111
Hawaii	✓	✓	✓		1
Idaho	✓	✓			47
Illinois	✓	✓	✓		186
Indiana	✓	✓	✓		28
Iowa	✓	✓	✓		41
Kansas	✓	✓		✓	56
Kentucky	✓	✓		✓	12
Louisiana	✓	✓	✓		28
Maine	✓	✓	✓		7
Maryland	✓	✓	✓	✓	81
Massachusetts	✓	✓	✓	✓	147
Michigan		✓			164
Minnesota	✓	✓	✓	✓	27
Mississippi	✓	✓	✓		130
Missouri		✓	✓	✓	44
Montana		✓		*	27
Nebraska		✓	✓		27
Nevada	✓	✓	✓		37
New Hampshire	✓	✓	✓	✓	18
New Jersey	✓	✓			80
New Mexico		✓	✓	*	78
New York	✓	✓			81
North Carolina	✓	✓	✓		84
North Dakota		✓			6
Ohio	✓	✓		✓	108
Oklahoma	✓	✓	✓		85
Oregon	✓	✓			164
Pennsylvania		✓			58
Rhode Island	✓	✓	✓	✓	18
South Carolina	✓	✓	✓		129
South Dakota		✓	✓	✓	7
Tennessee	✓	✓		✓	49
Texas	✓	✓	✓		84
Utah	✓	✓	✓		31
Vermont	✓	✓	✓		20
Virginia	✓	✓	✓	✓	54
Washington	✓	✓	✓	✓	31
West Virginia		✓			32
Wisconsin	✓	✓	✓		59
Wyoming	✓	✓	✓	✓	6
# of states responding yes	38	49	32	17	∅
Mean/Average	Yes	Yes	Yes	No	71

Legend:
 ✓ = Yes
 ∅ = Not applicable
 * = Officials could not access data

Sources: State surveys 2010/2011, LAB 2011, National Center for Safe Routes to School 2011

Bicycle Promotion in Cities

84% of cities report having Bike to Work Day events.

City	Bicycle education courses		Bike to Work Day events	Open street initiatives ⁽⁴⁾	City-sponsored bicycle ride	Public bike share program
	Youth	Adult				
Albuquerque	✓★	✓★	✓★	✓★		(1)
Arlington, TX	✓★ ⁽³⁾	✓★ ⁽³⁾	✓★			
Atlanta	(2)	✓	✓	✓★		
Austin	✓★	✓	✓	(1)	*	(1)
Baltimore	(1,2)	(1)	✓	✓	✓	(1)
Boston	✓	✓	✓	✓	✓	(1)
Charlotte	✓	✓	✓		✓	
Chicago	✓	✓★	✓	(5)	✓★	✓
Colorado Springs	*	✓	✓		✓★	
Columbus	✓	✓	✓		✓	
Dallas	✓★	✓			✓	(1)
Denver	✓	✓★	✓	(1)	✓	✓
El Paso				✓★	✓★	(1)
Fort Worth	✓★		✓★			(1)
Fresno	✓	✓★	✓		✓★	
Honolulu	✓	✓	✓	✓	✓	(1)
Houston	✓	✓	✓		✓	(1)
Jacksonville	✓★	✓★	✓★			
Kansas City, MO	✓	✓	✓	✓	✓★	(1)
Las Vegas	✓★					
Long Beach	✓	✓	✓★	✓	✓	(1)
Los Angeles	✓★	✓	✓	✓★		(1)
Louisville	✓	✓	✓	✓★	✓	(1)
Memphis	✓★	✓	✓★		✓	
Mesa	✓		✓		✓	(1)
Miami		✓	*	✓	✓★	(1)
Milwaukee	✓	✓	✓★		*	
Minneapolis	✓	✓	✓		✓	✓
Nashville			✓		✓	✓
New Orleans	✓★	✓★				
New York	✓	✓	✓	✓	✓	(1)
Oakland	✓	✓	✓	✓★	✓★	
Oklahoma City		✓	✓			(1)
Omaha	✓	✓	✓		✓	(1)
Philadelphia	✓	✓	✓	✓	✓	(1)
Phoenix		✓	✓	✓	✓	(1)
Portland, OR	✓	✓	✓	✓	✓	(1)
Raleigh			✓		✓★	(1)
Sacramento	✓★	✓★	✓★		✓★	(1)
San Antonio	✓★	✓★	✓★	(1)	✓★	(1)
San Diego	✓	✓	✓		*	(1)
San Francisco	✓	✓	✓	✓	✓	(1)
San Jose	✓	✓	✓	✓★	✓	(1)
Seattle	✓	✓	✓	✓		(1)
Tucson	✓	✓	✓	✓★		
Tulsa	✓★	✓★	✓	*	*	*
Virginia Beach		✓★	✓			
Washington, DC	✓	✓	✓	✓	✓	✓
# of cities responding yes	38	41	43	21	32	5
Mean/Average	Yes	Yes	Yes	No	Yes	No

Legend:

✓ = Yes
 ∅ = Not applicable
 * = Officials could not access data
 ★ = New since 2010
 Benchmarking Report

Source: City surveys 2010/2011 Notes: Cleveland, Detroit, and Indianapolis did not provide data requested for this chart. (1) In development. (2) Previous education program lost funding and agency is working on reinstating. (3) New in 2011. (4) Open streets are events where streets are temporarily closed to cars and are also known as "ciclovias," "Sunday parkways," and by other names. (5) Initiative no longer in operation or is currently inactive.

"Share the Road") campaign, whether states include driver's manual and driver's test information on bicyclists, and the number of schools participating in National Walk and Bike to School Day, and whether a state has a state-sponsored ride to promote bicycling or physical activity (and how many participants). City education benchmarks include the presence of youth and adult bicycle education courses and participation levels in these courses.

Share the Road Campaigns

"Share the Road" is perhaps the most common slogan used in bicycle education. Share the Road campaigns are widespread and can take many forms. Many states have Share the Road signs on roadways. Others have Share the Road bumper stickers. Some states have sophisticated campaigns with public service announcements including ads on buses, billboards, radio, and television. The basic message is always the same, encouraging bicyclists and motorists to obey traffic laws and show respect to other road users. Thirty-eight states report having a Share the Road or similar public safety campaign.

Driver Education

Driver education is a unique opportunity to instill knowledge about traffic laws and safety that individuals will use to form habits for years to come. The League of American Bicyclists' Bicycle Friendly State surveys collect information from states on whether information on bicycling is included in the state driver's manual and whether questions on sharing the roadway with bicyclists are included on the state driver's exam. Arkansas is the only state that does not include information on

bicycling in its state driver's manual. Thirty-two states include driver's test questions on bicyclists.

Bicycle Education

Although nearly everyone must have some form of driver's education before receiving a license, there is no education requirement to ride a bicycle. Yet having knowledge and skills to properly handle a bicycle in traffic can improve safety for bicyclists and make them better motorists. Bicycle education teaches youth and adults the rules of the road, how to properly handle a bicycle in traffic, and how to respectfully share the road with other users.

The Alliance survey on youth and adult bicycle education courses reveals that 41 cities (80% of cities surveyed, up from 35 cities 2 years ago) have adult bicycle education courses, and 38 cities

(Continued page 120)



Adult Bicycle Education Courses

City	Participation - # of adults					# adults per 1 participant				
	2006	2007	2008	2009	2010	2006	2007	2008	2009	2010
Atlanta	*	110	274	116	101	*	3,106	1,247	3,694	4,243
Austin	50	50	80	ø	108	11,545	11,545	7,215	ø	5,696
Baltimore	*	*	30	(1)	(1)	*	*	16,149	ø	ø
Boston	*	*	200	*	*	*	*	2,483	*	*
Colorado Springs	*	12	20	*	*	*	24,343	14,606	*	*
Chicago	*	*	*	*	200	*	*	*	*	10,910
Columbus	30	30	30	150	200	18,569	18,569	18,569	3,962	2,972
Dallas	-	66	124	300	300	*	13,716	7,300	3,188	3,188
Denver	*	*	*	39	64	*	*	*	12,057	7,347
Fresno	*	*	*	18	30	*	*	*	18,475	11,085
Houston	300	400	600	*	*	5,049	3,787	2,524	*	*
Indianapolis	*	*	30	*	*	*	-	19,297	*	*
Kansas City, MO	*	120	*	0	118	*	2,754	-	ø	3,126
Long Beach	*	16	70	*	*	*	20,910	4,779	*	*
Los Angeles	*	*	*	351	135	*	*	*	8,260	21,477
Louisville	60	120	60	50	50	7,111	3,556	7,111	8,659	8,659
Memphis	*	*	*	300	400	*	*	*	1,644	1,233
Miami	16	-	17	*	*	17,441	*	16,415	*	*
Milwaukee	10	20	25	100	200	41,919	20,959	16,768	4,436	2,218
Minneapolis	*	*	335	1,707	1,678	*	*	849	178	181
Nashville	30	30	30	(1)	(1)	15,229	15,229	15,229	ø	ø
New Orleans	*	*	*	0	20	*	*	*	ø	13,928
New York	101	429	945	1,307	1,785	63,083	14,852	6,742	4,974	3,642
Oakland	*	35	114	114	137	*	7,997	2,455	2,807	2336
Omaha	10	*	*	20	25	28,076	-	-	16,808	13,447
San Francisco	250	300	389	1,000	1,000	2,632	2,193	1,691	698	698
San Jose	200	200	200	250	200	3,505	3,505	3,505	2,878	3,598
Seattle	*	*	*	1,000	1,000	*	*	*	524	524
Tucson	400	400	500	562	*	1,000	1,000	800	738	*
Virginia Beach	*	*	*	0	20	*	*	*	ø	16,378
Washington, DC	*	*	*	434	893	*	*	*	1,120	544
Mean/Average	121	146	204	372	394	8,773 (2)	6,230 (2)	4,016 (2)	2,345 (2)	2,391 (2)
Median	55	88	97	150	169	13,387	9,771	6,927	3,441	3,620
High	400	429	945	1,707	1,785	63,083	24,343	19,297	18,475	21,477
Low	10	12	17	0	20	1,000	1,000	800	178	181

Source: City surveys 2008/2009 and 2010/2011 Notes: The following cities reported having adult bicycle education courses in at least one year, but did not provide data on participation: Albuquerque, Arlington, TX, Charlotte, Fort Worth, Honolulu, Jacksonville, Oklahoma City, Philadelphia, Phoenix, Portland, OR, Sacramento, San Antonio, San Diego, and Tulsa. All other cities not included in this table reported no adult bicycle education courses. (1) This city reports no longer having adult bicycle education courses. (2) Weighted average.

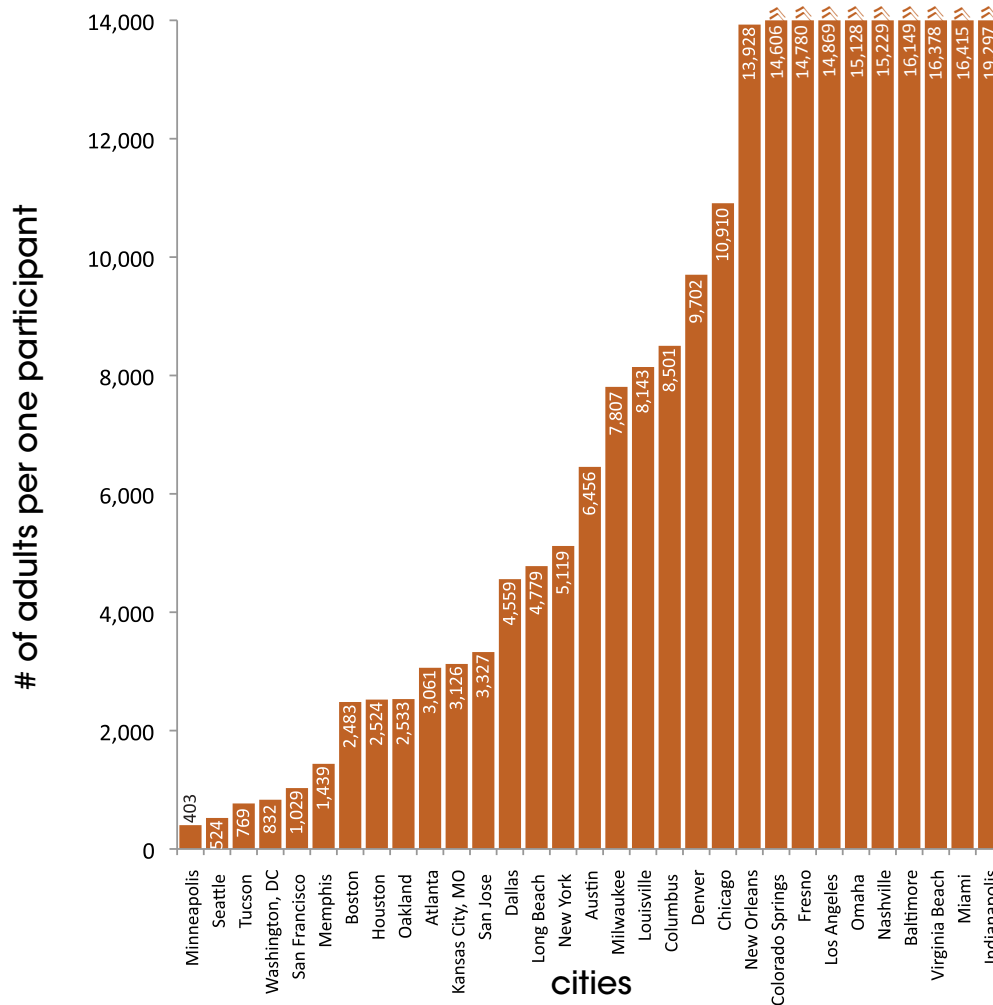
Legend:

- * = Officials could not access data
- ø = Not applicable
- = High value
- = Low value

Adult bicycle education participation has grown 267% in the last 4 years.

Forty-one of the cities surveyed for this report say their city has adult bicycle education courses. Since 2006, participation in these courses has been on the rise with the average number of participants increasing 267% in just four years. In 2010 these courses attracted an average of one person per 2,391 adults.

Adult Bicycle Education: Adults Per One Participant



Source: City surveys 2008/2009 and 2010/2011, ACS 2009 **Notes:** 3-year average data between 2008 and 2010 used with the following exceptions: 1-year data used for: Baltimore, Boston, Chicago, Colorado Springs, Houston, Indianapolis, Kansas City, MO, Long Beach, Miami, Nashville, New Orleans, Virginia Beach; 2-year average used for: Austin, Denver, Fresno, Los Angeles, Memphis, Omaha, Seattle, Tucson, Washington, DC. The following cities reported having adult bicycle education courses in at least one year, but did not provide data on participation: Albuquerque, Arlington, Charlotte, Fort Worth, Honolulu, Jacksonville, Oklahoma City, Philadelphia, Phoenix, Portland, Sacramento, San Antonio, San Diego, Tulsa. All other cities not included in this table reported no adult bicycle education courses.

One out of every 403 Minneapolis adults was a bicycle education participant, on average, between 2008-2010. Seattle, Tucson, and Washington, DC, also rank among the cities with the highest levels of participation in adult bicycle education courses.

Minneapolis leads cities for adult bicycle education participation.

Youth₍₃₎ Bicycle Education Courses

City	Participation - # of youth					# of youth per one participant				
	2006	2007	2008	2009	2010	2006	2007	2008	2009	2010
Albuquerque	*	*	*	10,000	10,000	*	*	*	13	13
Atlanta	*	300	34	(1)	(1)	*	303	2,671	ø	ø
Austin	*	*	*	1,500	1,000	*	*	*	117	175
Boston	*	*	300	800	2,600	*	*	388	139	43
Chicago	*	*	*	*	1,200	*	*	*	*	557
Columbus	3,800	4,200	4,500	4,000	4,000	46	42	39	45	45
Dallas	*	*	*	15	30	*	*	*	22,879	11,439
Honolulu	5,000	5,000	5,000	5,000	5,000	12	12	12	13	13
Houston	*	*	2,050	*	*	*	*	260	*	*
Long Beach	ø	ø	2,070	*	*	ø	ø	60	*	*
Los Angeles	*	*	*	7,851	2,427	*	*	*	119	384
Louisville	30	60	60	0	25	4,491	2,246	2,246	ø	5,341
Memphis	*	*	*	75	100	*	*	*	2,445	1,834
Mesa	*	*	*	200	600	*	*	*	611	204
Milwaukee	500	800	1,025	4,500	5,000	326	204	159	36	32
Minneapolis	125	150	180	1,126	729	534	445	371	72	112
Nashville	250	500	1,000	(1)	(1)	546	273	136	ø	ø
New Orleans	*	*	*	0	50	*	*	*	ø	1,526
New York	*	658	857	4,410	7,241	ø	2,892	2,221	429	261
Oakland	520	673	613	983	1,469	152	117	129	91	61
Omaha	1,000	1,000	1,000	(1)	(1)	94	94	94	ø	ø
Portland, OR	*	*	*	2,558	1,315	*	*	*	44	85
Sacramento	*	*	*	4,000	4,000	*	*	*	30	30
San Francisco	*	2000	3000	*	*	*	54	36	*	*
San Jose	25,000	25,000	25,000	22,008	19,967	9	9	9	11	12
Seattle	*	*	*	19,500	20,600	*	*	*	5	4
Tucson	*	*	*	875	*	*	*	*	148	*
Washington, DC	*	*	*	5,608	2,389	*	*	*	20	48
Mean/Average	4,466	3,337	2,917	4,524	4,273	31 (2)	80 (2)	88 (2)	58 (2)	67 (2)
Median	520	737	1,000	2,558	2,389	152	161	148	72	85
High	25,000	25,000	25,000	22,008	20,600	4,491	2,892	2,671	22,289	11,439
Low	30	60	34	0	25	9	9	9	5	4

Source: City surveys 2008/2009 and 2010/2011 **Notes:** The following cities reported having youth bicycle education courses in at least one year, but did not provide data on participation: Arlington, TX, Baltimore, Charlotte, Denver, Fort Worth, Fresno, Indianapolis, Jacksonville, Kansas City, MO, Las Vegas, Philadelphia, San Antonio, San Diego, Tulsa, and Virginia Beach. All other cities not included in this table reported no youth bicycle education courses. (1) This city reports no longer having youth bicycle education courses. (2) Weighted average. (3) "Youth" includes all residents under age 18.

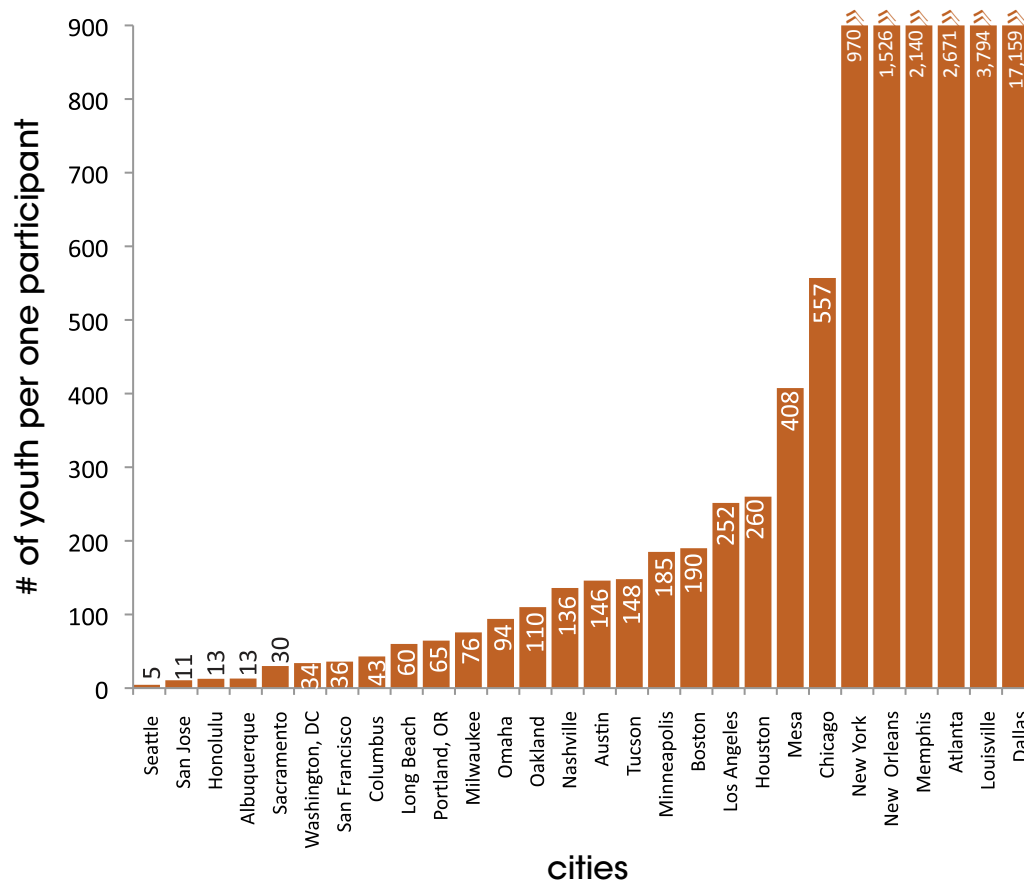
Legend:

- * = Officials could not access this data
- ø = Not applicable
- Dark Blue = High value
- Light Blue = Low value

One of every 67
youth participate
in bicycle
education
courses in major
U.S. cities.

Thirty-seven of the cities surveyed for this report have youth bicycle education courses. On average, 1 out of 67 youth (under age 18) attend a youth bicycle education course in these cities (in 2010). Seattle has the high value for youth bicycle participation in 2010 with 20,600 participants. One of every four Seattle youth participated in bicycle education in 2010.

Youth⁽³⁾ Bicycle Education: Youth per One Participant



Sources: City surveys 2008/2009 and 2010/2011, ACS 2009 **Notes:** 3-year average data between 2008 and 2010 used with the following exceptions: 1-year data used for: Atlanta, Chicago, Houston, Long Beach, Nashville, Omaha, San Francisco, and Tucson; 2-year average used for: Albuquerque, Austin, Dallas, Los Angeles, Louisville, Memphis, Mesa, Oakland, Portland, OR, Sacramento, Seattle, and Washington, DC. The following cities reported having youth bicycle education courses in at least one year, but did not provide data on participation: Arlington, Baltimore, Charlotte, Denver, Fort Worth, Fresno, Indianapolis, Jacksonville, Kansas City, MO, Las Vegas, Philadelphia, San Antonio, San Diego, Tulsa, and Virginia Beach. All other cities not included in this table reported no youth bicycle education courses. (3) "Youth" includes all residents under age 18.

One out of every five youth in Seattle participate in bicycle education courses. San Jose, Honolulu, Albuquerque, and Sacramento also lead other major U.S. cities in participation levels for youth bicycle education courses.

**Seattle now leads
cities for youth
bicycle education
participation.**



(75% of cities surveyed, up from 30 cities 2 years ago) have youth bicycle education courses. These education courses vary in that some are sponsored by the local government, some by a local nonprofit or advocacy organization, some by the local police department, and others are the result of partnerships between multiple agencies. Surveys also reveal that city adult bicycle education courses averaged one participant per 2,391 adults, and youth courses average one participant per 67 youth residents (in 2010).

Encouragement Programs and Events

Encouragement programs are those activities which support and promote bicycling and walking. There are many different types of encouragement activities, but this report looked at four specific types of common encouragement events: Bike to Work Day, Walk and Bike to School Day, city-sponsored bicycle rides, and open streets (ciclovía) initiatives. This report also looked at participation levels of these efforts to establish benchmarks and baseline data to measure progress among cities going forward.

Bike to Work Day

Bike to Work Day is an annual event held on the third Friday in May throughout most of the United States and Canada. Since the League of American Bicyclists organized the first Bike to Work Day in 1956, the day has been a rallying point for bicycle advocates to promote bicycling as a healthy and fun

Bike to Work Day Events

City	Participation - # of adults					# of adults per one participant				
	2006	2007	2008	2009	2010	2006	2007	2008	2009	2010
Albuquerque	Ø	Ø	Ø	800	800	Ø	Ø	Ø	501	501
Arlington, TX	Ø	Ø	Ø	*	*	Ø	Ø	Ø	*	*
Atlanta	*	*	100	341	251	*	*	3,417	1,257	1,707
Austin	*	50	165	*	1,133	*	11,545	3,498	*	543
Baltimore	*	250	250	186	220	*	1,938	1,938	2,658	2,248
Boston	*	*	3,000	3,000	3,000	*	*	166	178	178
Charlotte	*	*	*	*	*	*	*	*	*	*
Chicago	2,000	2,000	2,000	4,000	5,000	1,027	1,027	1,027	545	436
Colorado Springs	900	1,000	1,200	*	*	325	292	243	*	*
Columbus	50	50	450	400	300	11,141	11,141	1,238	1,486	1,981
Denver	20,500	21,000	35,000	23,284	17,093	22	21	13	20	28
Detroit	75	75	75	*	*	7,652	7,652	7,652	*	*
Fort Worth	Ø	Ø	Ø	*	80	Ø	Ø	Ø	*	6,424
Fresno	200	350	400	400	570	1,623	927	812	831	583
Honolulu	200	200	200	200	200	1,464	1,464	1,464	1,557	1,557
Houston	276	321	350	340	225	5,488	4,718	4,328	4,866	7,354
Indianapolis	*	435	500	*	*	*	1,331	1,158	*	*
Jacksonville	Ø	Ø	Ø	100	100	Ø	Ø	Ø	6,117	6,117
Kansas City, MO	250	500	1,100	*	1,421	1,322	661	300	*	260
Long Beach	*	*	150	*	*	*	*	2,230	*	*
Los Angeles	*	*	*	*	4,500	*	*	*	*	644
Louisville	150	200	150	600	550	2,844	2,133	2,844	722	787
Memphis	Ø	Ø	Ø	0	150	Ø	Ø	Ø	Ø	3,289
Mesa	*	*	25	7	77	*	*	14,382	49,282	4,480
Miami	*	*	*	*	*	*	*	*	*	*
Milwaukee	Ø	Ø	Ø	*	*	Ø	Ø	Ø	*	*
Minneapolis	300	400	2,500	2,526	7,295	948	711	114	120	42
Nashville	50	50	100	*	*	9,137	9,137	4,569	*	*
New York	*	*	750 (1)	*	*	*	*	8,495	*	*
Oakland	1,294	1,401	1,732	3,861	4,000	216	200	162	83	80
Oklahoma City	200	200	200	130	130	2,024	2,024	2,024	3,217	3,217
Omaha	330	400	600	900	1,000	851	702	468	374	336
Philadelphia	200	300	300	*	*	5,436	3,624	3,624	*	*
Phoenix	25	25	75	75	90	42,991	42,991	14,330	15,029	12,524
Portland, OR	*	*	*	11,350	11,050	*	*	*	40	41
Raleigh	50	63	75	50	75	5,454	4,329	3,636	6,160	4,107
Sacramento	Ø	Ø	Ø	*	*	Ø	Ø	Ø	*	*
San Antonio	Ø	Ø	Ø	1,000	*	Ø	Ø	Ø	991	*
San Diego	*	*	*	*	*	*	*	*	*	*
San Francisco	*	65,000	75,000	*	*	*	10	9	*	*
San Jose	13,000	14,000	15,000	*	*	54	50	47	*	*
Seattle	*	*	*	19,097	14,250	*	*	*	27	37
Tucson	300	400	500	*	*	1,333	1,000	800	*	*
Tulsa	*	*	*	*	*	*	*	*	*	*
Virginia Beach	*	*	*	*	*	*	*	*	*	*
Washington, DC	*	*	*	8,127	9,184	*	*	*	60	53
Mean/Average	2,018	4,528	4,895	500	3,065	300 (2)	132 (2)	157 (2)	238 (2)	286 (2)
Median	225	336	400	3,366	570	1,544	1,398	1,464	831	644
High	20,500	65,000	75,000	23,284	17,093	42,991	42,991	14,382	49,282	12,524
Low	25	25	25	0	75	22	10	9	20	28

Source: City surveys 2008/2009 and 2010/2011 **Notes:** The following cities did not report having Bike to Work Day events and are not included in this table: Dallas, El Paso, and New Orleans. Las Vegas previously reported having a Bike to Work Day event and now reports having no Bike to Work Day events. Cleveland, Detroit, and Indianapolis did not respond to requests for data from the 2010/2011 survey. (1) Answered "700-800." (2) Weighted average.

Legend:

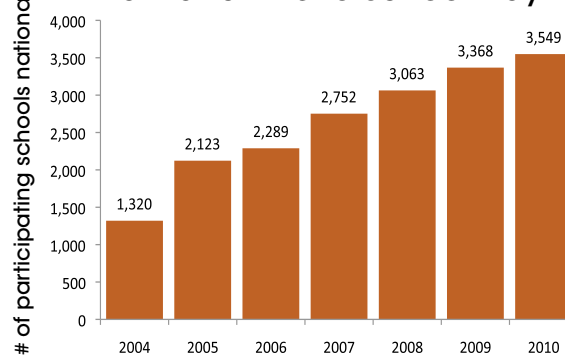
* = Officials could not access data
 Ø = Not applicable
 ■ = High value ■ = Low value



Former Congressman Jim Oberstar, originator of the national Safe Routes to School Program, visits students who bike and walk to school at Beach Elementary in Portland, OR. Photo courtesy of Greg Raisman

alternative to driving. Local advocacy organizations and government agencies across North America organize bicycling encouragement and promotion events around Bike to Work Day including commuter challenges, organized rides, energizer stations (with coffee, breakfast treats, and bicycling literature), and more. Bike to Work Day is the most common encouragement activity among major U.S. cities with 43 cities reporting some organized event around this day (up from 38 cities 2 years ago). Both government and nonprofit organizations sponsor these events. In 2010, cities averaged one Bike to Work Day participant for every 286 adults. Denver reports the greatest per capita participation with 1 out of every 28 adults, in 2010.

Schools Participating in National Walk and Bike to School Day

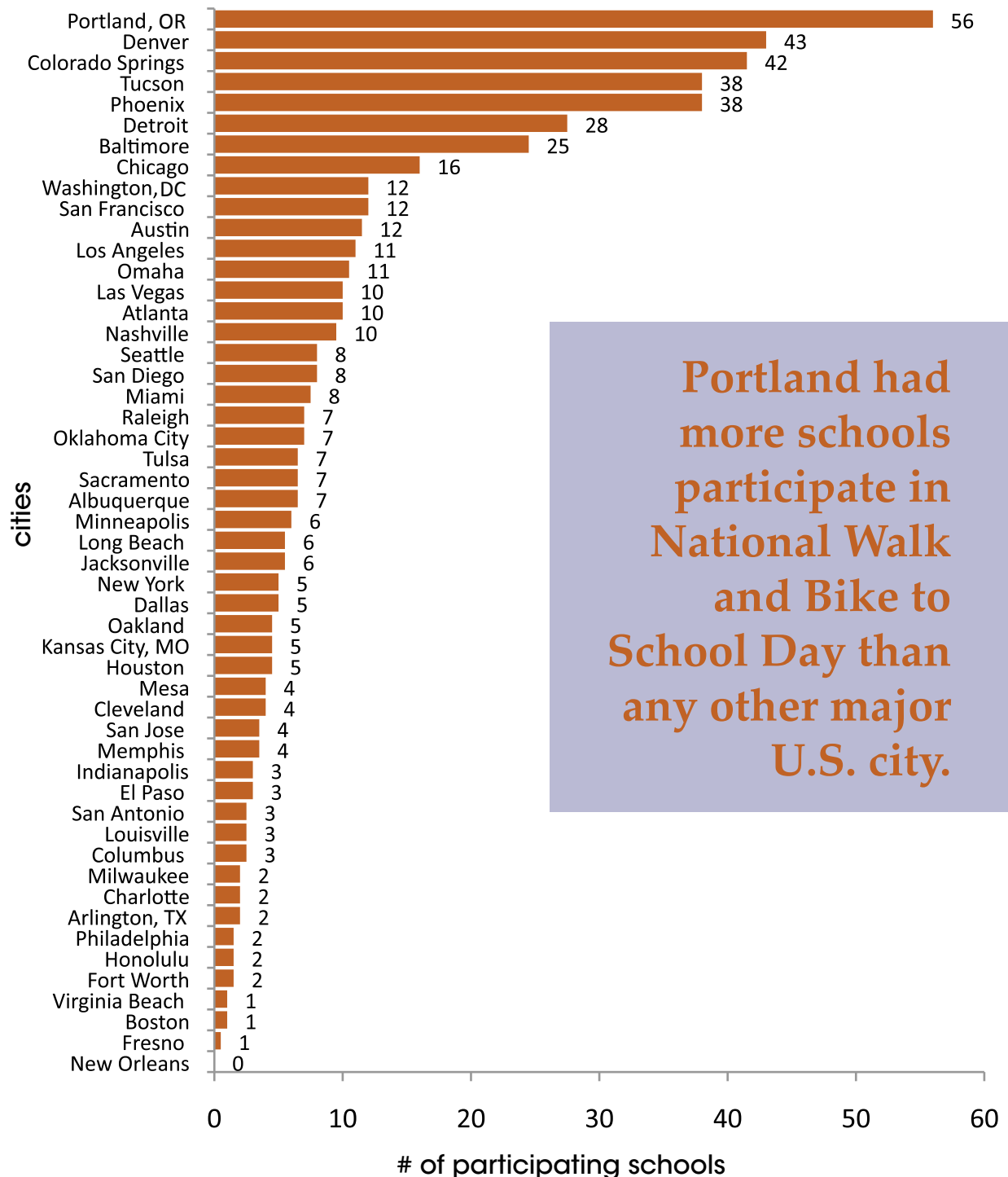


Source: NCSRTS 2011

Walk and Bike to School Day

Walk and Bike to School Day is an annual international event held the first Wednesday in October to promote and encourage kids bicycling and walking to school. The first

Number of Schools Participating in Walk and Bike to School Day (2-year average 2009-2010)



Source: NCSRTS 2011

Walk to School Day was in 1995 in Hertfordshire, England. By 2000 the first International Walk to School Day was held with events throughout Europe, Canada, and the United States. Communities can choose to celebrate International Walk to School Day for a day, a week, or an entire month. Events can range from simply encouraging parents and children to bicycle or walk to school to an organized walk or bicycle to school parade with refreshments and prizes for children who participate.

In 2010, over 3,500 schools from all 50 states participated in Walk to School Day. New Orleans was the only major U.S. city in this report that did not have any schools registered for Walk to School Day in 2009 or 2010. Portland, OR, and Denver had more schools registered for Walk to School Day than any other major U.S. cities with 56 and 43 registered schools, respectively.

Promotional Bicycle Rides

Promotional bicycle rides are another popular encouragement activity that many states, cities, and advocacy organizations sponsor. While temporary in nature, these events can promote bicycling as a healthy and fun way to get around, and can raise awareness around local bicycle routes, issues, and groups. They are often an excellent entry point for new bicyclists who are not yet comfortable riding alone in traffic, but who will try out bicycling in a group ride setting. Promotional rides are also great opportunities for media coverage and forging new partnerships between bicycling and walking organizations and other government and community groups.

Seventeen states and 32 cities report having government-sponsored rides to promote bicycling or physical activity. New York City's city-sponsored ride (TD Bank Five Boro Bike Tour) attracts 30,000 bicyclists, more than any other city-sponsored ride. Louisville's Mayor's Healthy Hometown Hike and Bike attracts one participant for every 79 residents, making it the highest per capita participation of any city-sponsored ride.

Open Street Initiatives

Although some cities like Seattle and San Francisco have had car-free public space for decades, a new sort of car-free event has been sweeping North American cities in the last several years. These events, often called "open streets," "Sunday parkways," or "ciclovias" are modeled after the successful Ciclovía program in Bogotá, Colombia. These events temporarily shut down a portion of roadways to cars to create a connected nonmotorized network for bicycling, walking, running, skating, and a number of other organized activities.

Twenty-one cities report having hosted an open streets event. New York's Summer Streets drew 195,000 people in 2010, more than any other ciclovía event. Portland and Seattle had the most participants per capita at their open streets events with one out of every six residents participating. These high participation levels for relatively new events demonstrate a large interest in, and latent demand for, safe places to bicycle and walk.

City-Sponsored Bicycle Rides

City	Participation - # of people					# of people per one participant				
	2006	2007	2008	2009	2010	2006	2007	2008	2009	2010
Baltimore	*	*	25/wk	2,000	3,000	*	*	25,498	319	212
Boston	*	*	600	4,000	4,000	*	*	1,022	161	161
Charlotte	*	*	*	300	300	*	*	*	2,348	2,348
Chicago	*	*	*	22,500	24,000	*	*	*	127	119
Columbus	∅	∅	200 (1)	250	1,000	∅	∅	3,665	3,092	773
Dallas	3,000	3,000	3,000	80	120	413	413	413	16,245	10,830
Denver	∅	∅	∅	∅	115	*	*	*	∅	5,307
Honolulu	4,000	3,500	3,400	3,000	3,000	88	101	104	125	125
Houston	*	2,500	3,758	3,000	2,300	*	819	545	754	983
Long Beach	*	*	4,000	*	1,000	*	*	115	*	463
Los Angeles	1,200	1,700	2,300	(3)	(3)	3,172	2,239	1,655	∅	∅
Louisville	6,000	8,000	10,000	5,000	7,200	94	70	56	113	79
Mesa	∅	∅	∅	80	80	*	*	*	5,840	5,840
Miami	*	*	*	150	150	*	*	*	2,888	2,888
Minneapolis	*	4,700	3,400	3,800	3,300	*	75	103	101	117
Nashville	1,000	1,000	1,500	*	*	593	593	396	*	*
New York	30,000	30,000	30,000	30,000	30,000	280	280	280	280	280
Oakland	*	*	*	50	50				8,183	8,183
Omaha	50	60	75	50	80	7,487	6,239	4,991	9,094	5,684
Philadelphia	*	2,400	3,100	*	450	*	604	468	*	3,438
Phoenix	924	912	1,023	764	526	1,725	1,747	1,558	2,086	3,030
Raleigh	∅	∅	∅	35	40	*	*	*	11,577	10,130
Sacramento	*	*	*	15	20	*	*	*	31,112	23,334
San Antonio	*	*	*	225		*	*	*	*	6105
San Francisco	2,000 (2)	2,000 (2)	2,000 (2)	(3)	(3)	382	382	382	∅	∅
San Jose	300	400	500	300	400	3,075	2,306	1,845	3,216	2,412
Washington, DC	∅	∅	3,000	*	*	∅	∅	196	*	*
Mean/Average	4,847	4,629	3,993	3,967	3,537	391 (4)	381 (4)	368 (4)	325 (4)	350 (4)
Median	1,600	2,400	2,650	300	450	503	593	441	2,348	2,412
High	30,000	30,000	30,000	30,000	30,000	7,487	6,239	25,498	31,112	23,334
Low	50	60	25	15	20	88	70	56	101	79

Legend:

- * = Officials could not access data
- ∅ = Not applicable
- = High value
- = Low value

Source: City surveys 2008/2009 and 2010/2011 Notes: The following cities reported having a city-sponsored bike ride in at least one year, but did not provide data on participation: Colorado Springs, El Paso, Fresno, Kansas City, and Portland. (1) First year of this event. (2) Answered "thousands," represented as 2,000. (3) This city reports no longer having a city-sponsored bicycle ride. (4) Weighted average.

Louisville's Mayor's Healthy Hometown Hike and Bike.
Photo courtesy of Louisville Metro Government



City-sponsored
bicycle rides
averaged one
participant per 350
residents in 2010.

Open Street/Ciclovia Initiatives

City	Participation - # of people					# of people per one participant				
	2006	2007	2008	2009	2010	2006	2007	2008	2009	2010
Atlanta	Ø	Ø	Ø	Ø	10,500 ⁽¹⁾	Ø	Ø	Ø	Ø	52
Baltimore	Ø	Ø	2,000 ⁽¹⁾	Ø ⁽⁵⁾	Ø ⁽⁵⁾	Ø	Ø	319	Ø	Ø
Boston	2,000	3,000	4,000	Ø ⁽⁵⁾	Ø ⁽⁵⁾	307	204	153	Ø	Ø
Kansas City, MO	Ø	Ø	3,600 ⁽¹⁾	Ø	Ø	Ø	Ø	122	Ø	Ø
Los Angeles	Ø	Ø	Ø	Ø	60,000	Ø	Ø	Ø	Ø	64
Louisville	Ø	Ø	Ø	40	50	Ø	Ø	Ø	14,162	11,330
Miami	Ø	Ø	2,800	2,000	5,000	Ø	Ø	125	217	87
New York	Ø	250 ^{(1) (2)}	150,000	205,000	195,000	Ø	33,098	55	41	43
Oakland	Ø	Ø	Ø	Ø	4,000	Ø	Ø	Ø	Ø	103
Phoenix	*	*	75,000	Ø	Ø	*	*	20	Ø	Ø
Portland, OR	Ø	Ø	15,000 ⁽¹⁾	62,000	91,000	Ø	Ø	37	9	6
San Francisco	Ø	Ø	7,000 ^{(1) (3)}	*	*	Ø	Ø	109	*	*
San Jose	Ø	Ø	Ø	Ø	5,000	Ø	Ø	Ø	Ø	193
Seattle	Ø	Ø	Ø	96,000	96,000	Ø	Ø	Ø	6	6
Tucson	Ø	Ø	Ø	Ø	5,000	Ø	Ø	Ø	Ø	109
Washington, DC	Ø	Ø	1,200	*	Ø	Ø	Ø	490	*	Ø
Mean/Average	Ø	Ø	28,9556	52,149	42,868	Ø	Ø	53 ⁽⁴⁾	41	37
Median	Ø	Ø	4,000	2,000	5,000	Ø	Ø	116	41	75.5
High	Ø	3,000	150,000	205,000	195,000	Ø	33,098	490	14,162	11,330
Low	Ø	250	1,200	40	50	Ø	204	20	6	6

Source: City surveys 2008/2009 and 2010/2011 Notes: Albuquerque, Chicago, El Paso, Honolulu, Houston, Long Beach, Philadelphia, and Seattle reported having an open street/ciclovia event during at least one of these years but could not access data on participation; all other cities did not report having an open street/ciclovia event. (1) First year of this event. (2) Answered "200-300." (3) Answered "several thousand." (4) Weighted average. (5) Report no longer having a ciclovia event.

Legend:

* = Officials could not access data
 Ø = Not applicable
 = High value
 = Low value

San Francisco children enjoy a street free of car traffic at one of the city's Sunday Streets events. Photo by Frank Chan, San Francisco Bicycle Coalition





Photo by Gabriella Salary

6: GRASSROOTS ADVOCACY

Growing the Movement

There is no doubt that bicycle and pedestrian advocacy is on the rise. When the Alliance for Biking & Walking was formed in 1996 as the North American coalition of grassroots bicycle and pedestrian advocacy organizations, there were just 12 member organizations. Today the Alliance includes over 185 organizations in 48 U.S. states, four Canadian provinces, and Mexico City. The number of bicycle and pedestrian advocacy organizations has steadily increased through the Alliance's comprehensive organizational development efforts and in response to increasing traffic congestion, rising gas prices, safety risks, a growing obesity epidemic, and climate change.

This upward trend in bicycle and pedestrian advocacy doesn't seem to be waning anytime soon. As these issues become more prevalent in mainstream public discourse, the Alliance is working with more emerging organizations and connecting their leaders with peers around the country as they embark to transform their communities into more vibrant, healthy, and livable places.

Advocacy as an Indicator

The presence and strength of advocacy organizations in states and cities have been used as indicators to measure the state of bicycling and walking. Strong advocacy organizations are often necessary to local jurisdictions with hopes of

passing and implementing progressive policies for bicycling and walking. Government and elected officials passionate about these issues often promote or work with emerging advocates, recognizing the need for increased citizen involvement in the public policy discourse.

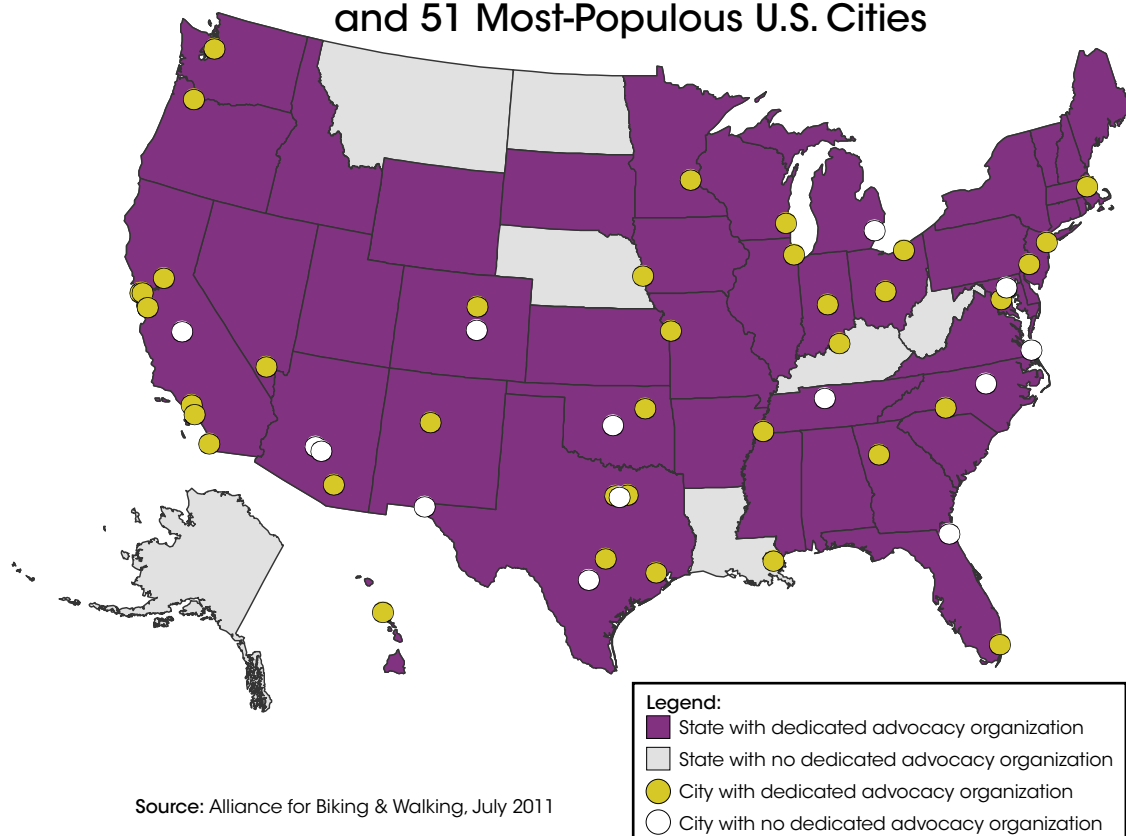
The breadth of missions encompassed by advocacy organizations that promote bicycling and walking is large. Among others, organizations may include urban bicycle advocacy, recreational bicycle clubs, bicycle sharing or community bike groups, and lobbying organizations. The presence and capacity of these organizations are both indicators of the prominence of bicycling and walking in communities and

are worth comparing to other bicycling and walking indicators. For this report, the Alliance for Biking & Walking only collected data from Alliance member organizations.

Proving Effectiveness

The Alliance's coalition of grassroots advocacy organizations is constantly influencing public policy and helping to create more bikeable and walkable communities. But can this be proven? Looking at data on capacity of advocacy organizations (page 153) shows there may be a connection between the advocacy capacity and bicycling and walking levels. Besides using these data to illustrate their effectiveness, Alliance leaders can also learn where they are

Alliance Bike/Ped Advocacy Organizations in 50 States and 51 Most-Populous U.S. Cities



Source: Alliance for Biking & Walking, July 2011

successful and which areas need greater attention, thus refocusing limited resources for the greatest impact.

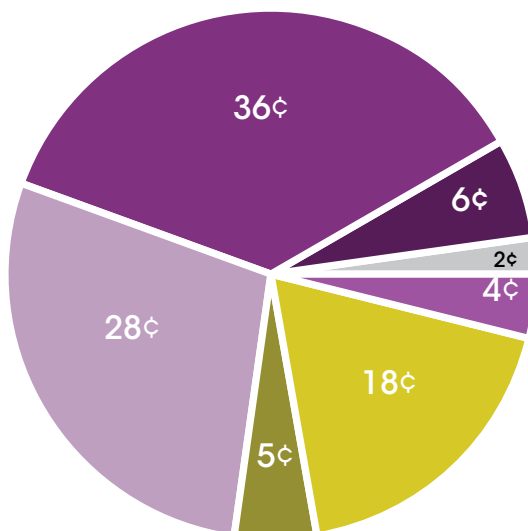
Measuring Advocacy Capacity

Measuring the capacity of advocacy organizations is not an easy thing to do. The Alliance collects data from member organizations annually as part of an "organization profile." Variables measured include revenue and revenue sources, membership, staffing, number of contacts and media impressions (among others). While these variables are telling in some cases, they will not accurately reflect the full capacity of these organizations. Some organizations with strong

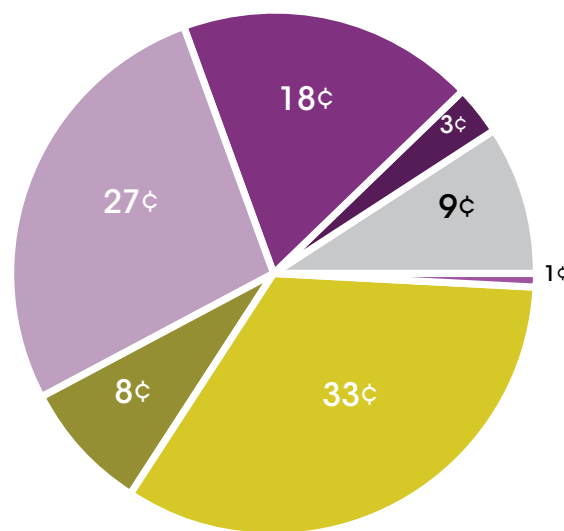
leaders and a dedicated base, can and do win great victories for bicycling and walking with virtually no budget. However, in the Alliance's experience, organizations with sustainable revenue sources and budgets to employ full-time staff are the most self-sustaining and are able to accomplish more in the long term. Membership numbers and media impressions were included because they increase the political influence of an organization.

Organizations are represented in this section by the state or city they serve. In the cases where more than one advocacy organization serves a particular state or city, both organizations are represented separately by their city name and a number. Appendix 2 contains the list of 50 states and 51 cities studied in

Breakdown of Every Dollar Earned by Statewide Alliance Organizations



Breakdown of Every Dollar Earned by Alliance Organizations Serving Cities



Legend:

- = Memberships and donations
- = Events, rides, galas, sponsorships
- = Fee-for-service activities
- = Foundation grants
- = Other
- = Government grants and contracts
- = Bicycle shops and manufacturers

(Graphs this page) Source: Alliance for Biking & Walking 2011 Note: Graphs consider only Alliance organizations serving states and one of the 51 study-area cities. To see what organizations are included, reference Appendix 2, page 200. Some values round down which makes total appear to be 99%.

Advocacy Capacity **RANKING**

States⁽¹⁾

1. Maine
2. Hawaii
3. Wyoming
4. Oregon
5. Colorado
6. Washington
7. Iowa
8. Minnesota
9. Connecticut
10. Michigan

11. Illinois
12. Massachusetts
13. Maryland
14. Indiana
15. New Mexico
16. Georgia
17. South Carolina
18. Arizona
19. Florida
20. Arkansas
21. New Hampshire
22. Mississippi
23. Missouri
24. Idaho
25. Utah
26. Pennsylvania
27. Wisconsin
28. New Jersey
29. New York
30. Alabama
31. California
32. Oklahoma
33. South Dakota
34. Delaware
35. Tennessee
36. Nevada
37. North Carolina

Cities⁽¹⁾

1. Seattle
2. San Francisco
3. Minneapolis
4. Boston
5. Chicago
6. New York
7. Honolulu
8. Oakland
9. Portland
10. Sacramento

11. Philadelphia
12. Tulsa
13. Indianapolis
14. Atlanta
15. Columbus
16. San Diego
17. New Orleans
18. Louisville
19. Miami
20. Memphis
21. Los Angeles/
Long Beach
22. Cleveland
23. Denver
24. Charlotte
25. Austin
26. Kansas City, MO
27. Milwaukee
28. Dallas/Fort Worth
29. Houston

this report and identifies the advocacy organization(s) representing each city or state. To see which organization(s) are represented by these data, you can cross-reference the city or state with Appendix 2, page 200.

The data in this chapter measure the capacity of Alliance member organizations only. Although most state and local bicycle and pedestrian advocacy organizations are Alliance members, a few are not. Also, many other organizations, individuals, and government officials advocate for bicycling and walking. This section is by no means an all-inclusive or definitive measurement of advocacy capacity across the United States. In the scope of this report, measurements are limited to the capacity of Alliance member state and local bicycle and pedestrian advocacy organizations.

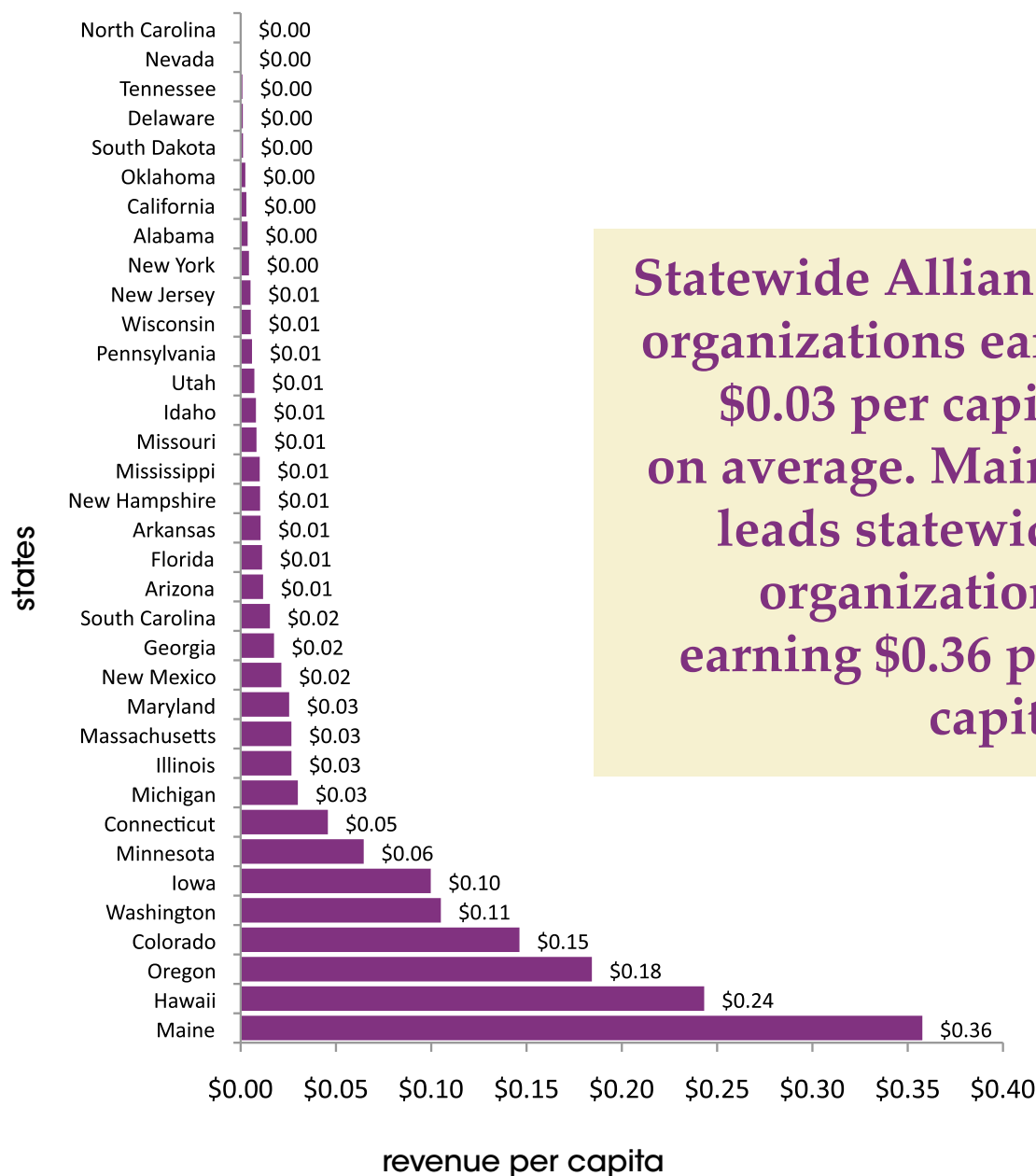
Revenue

The Alliance for Biking & Walking asked its member organizations for their revenue sources from the most re-
(Continued page 134)

Source: Alliance for Biking & Walking 2011 **Notes:** The rankings in the chart above are based upon the 2010 per capita income of advocacy organizations. States and cities represented by the advocacy organization with the highest per capita revenue are ranked #1 and so forth; Indiana and Michigan are ranked according to 2009 per capita revenue because 2010 data were unavailable. (1) To view which states and cities were not served by a dedicated Alliance advocacy organization and which are served by an Alliance advocacy organization that did not provide data on revenue for this ranking, reference Appendix 2, page 200.

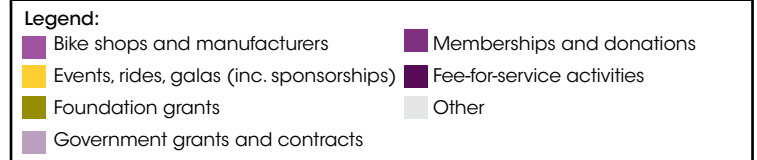
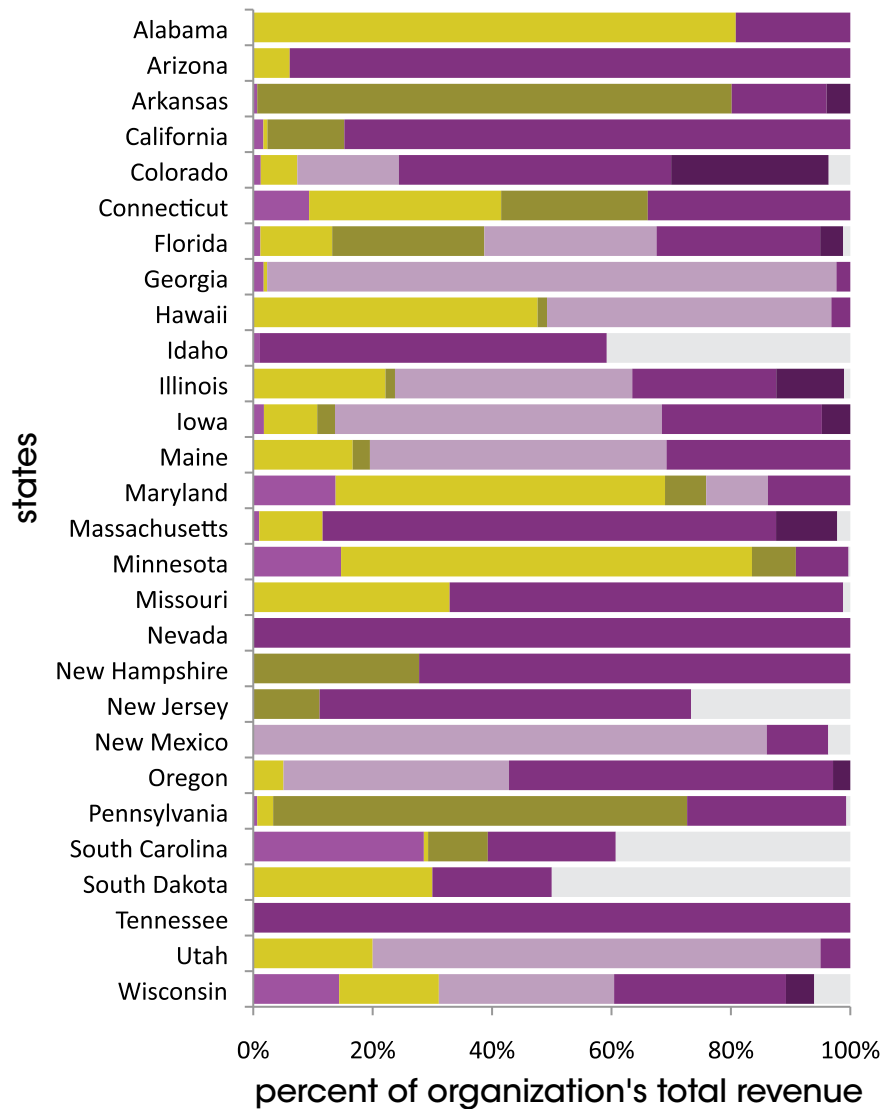


Per Capita Revenue of Statewide Alliance Organizations



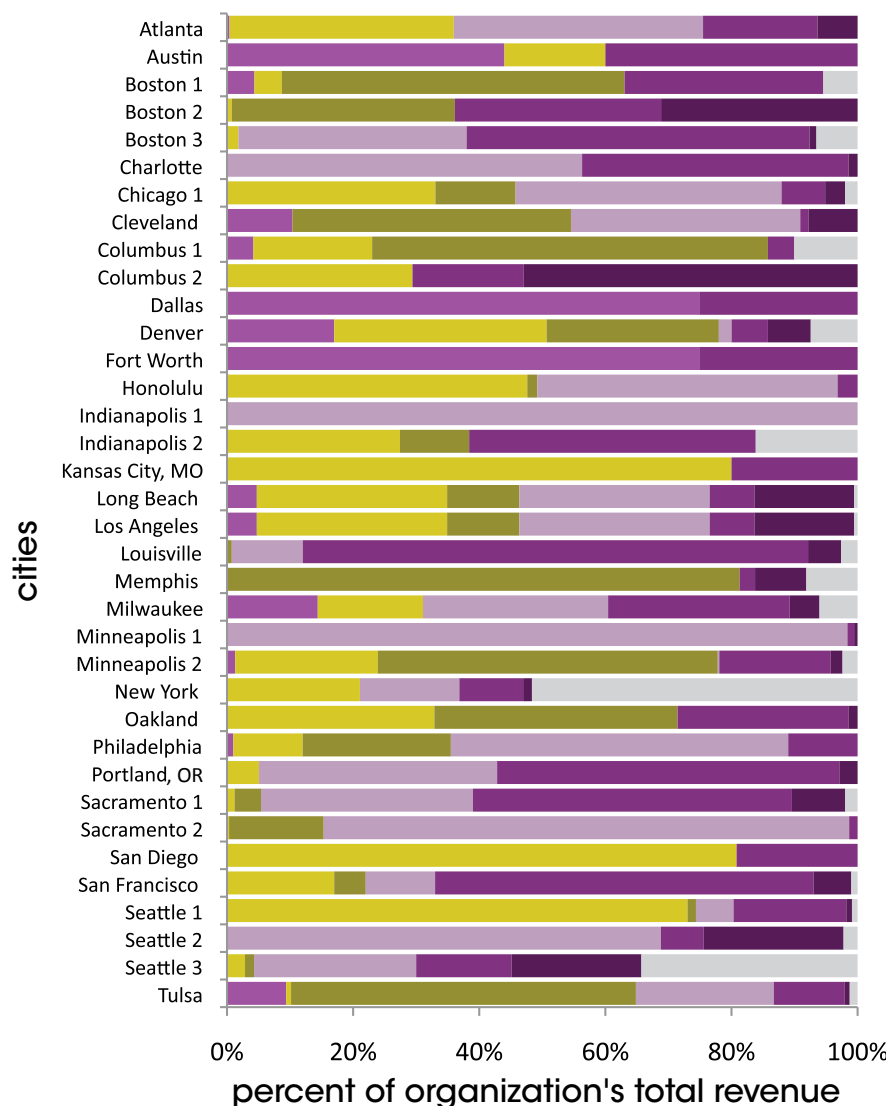
Source: Alliance for Biking & Walking 2011 **Notes:** Graph considers only Alliance organizations serving states. As of May 2011, the following states were not served by a dedicated statewide Alliance advocacy organization: Alaska, Kentucky, Louisiana, Montana, Nebraska, North Dakota, and West Virginia. The following states are served by a statewide Alliance advocacy organization that did not provide information on organization revenue: Indiana, Kansas, Ohio, Rhode Island, Texas, and Vermont; Data for Alabama, Georgia, Idaho, and New Hampshire are from 2009 because no 2010 data were available. To see the organization and study area matches, please reference Appendix 2, page 200.

Revenue Sources of Statewide Alliance Organizations



Source: Alliance for Biking & Walking 2011 **Notes:** Graph considers only Alliance organizations serving states. As of May 2011, the following states were not served by a dedicated statewide Alliance advocacy organization: Alaska, Kentucky, Louisiana, Montana, Nebraska, North Dakota, and West Virginia. The following states are served by a statewide Alliance advocacy organization that did not provide information on organization revenue: Indiana, Kansas, Ohio, Rhode Island, Texas, Vermont, and Wyoming; Data for Alabama, Georgia, Idaho, and New Hampshire are from 2009 because no 2010 data were available. To see the organization and study area matches, please reference Appendix 2, page 200.

Revenue Sources of Alliance Organizations Serving Cities



Source: Alliance for Biking & Walking 2011 **Notes:** All data in this chart represent 2010 figures unless otherwise noted. As of May 2011, the following cities were not served by a dedicated local Alliance advocacy organization: Arlington, TX, Baltimore, Colorado Springs, Detroit, El Paso, Fresno, Jacksonville, Mesa, Nashville, Oklahoma City, Phoenix, Raleigh, San Antonio, and Virginia Beach. The following city are served by a local Alliance advocacy organization that did not provide data for this illustration: Albuquerque, Las Vegas, Miami, New Orleans, Omaha, San Jose, Tucson, and Washington, DC. Cities with numbers following indicate there is more than one advocacy organization serving this city. To see the organization and study area matches, please reference Appendix 2, page 200.

cent budget year. Data indicate that on average statewide advocacy organizations operate on three cents per capita. The state with the highest per capita income for an advocacy organization is Maine with a budget of 36 cents per capita.

Organizations that represent cities have significantly higher incomes per capita. On average, organizations representing cities earn 15 cents per capita. Seattle ranks highest in per capita earnings of all cities surveyed at \$1.99 per capita (combining the revenue of three local advocacy organizations).

The range is wide among states and cities in part because some advocacy

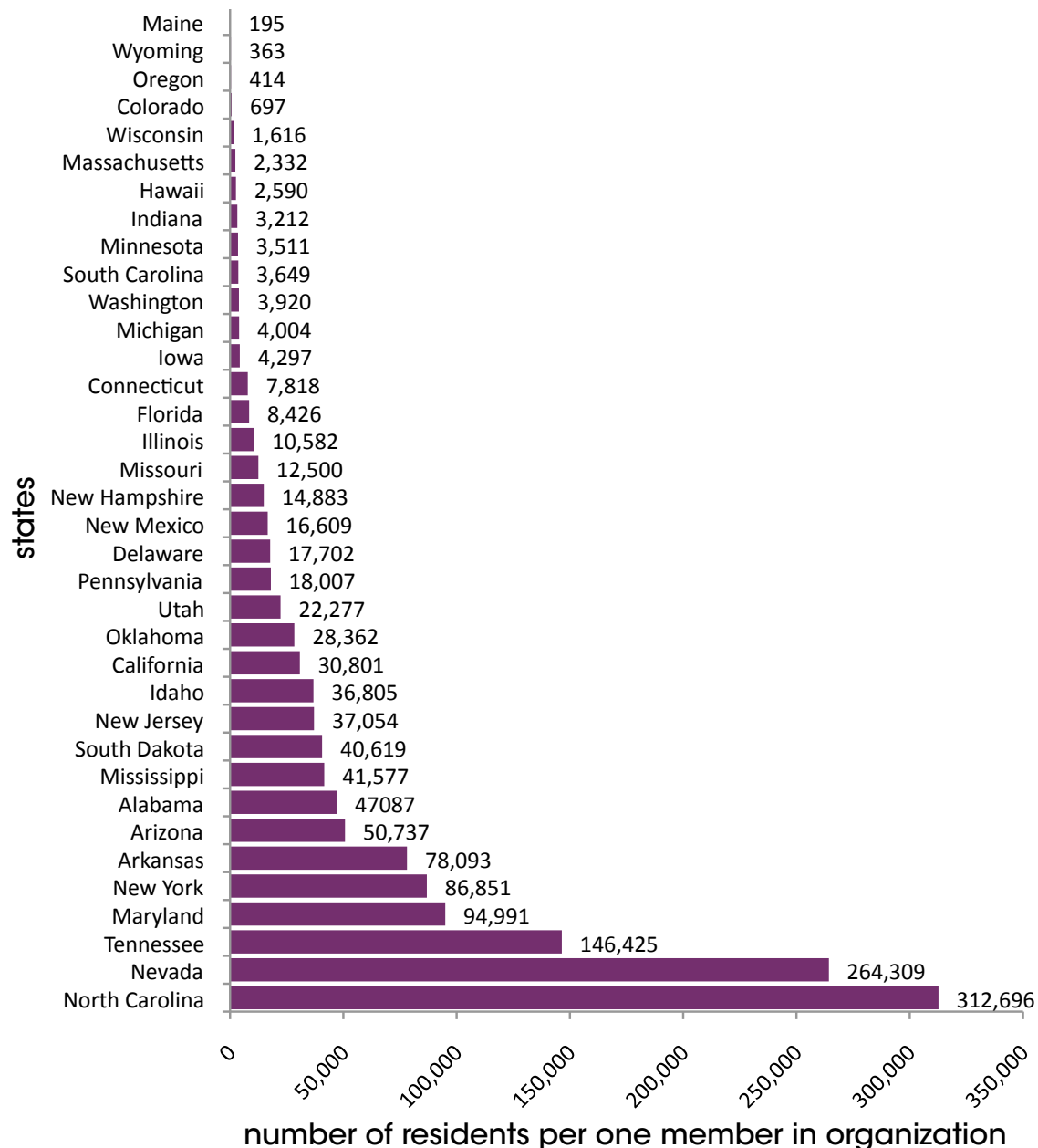
organizations are new and are being compared to longer-established organizations. Also, some organizations have full-time staff for fundraising while others are run by volunteers.

Both city and state organizations have diversified revenue mainly divided between memberships and donations, program fees and events, and government grants and contracts. Memberships and donations make up over a third of the revenue of statewide organizations. Government grants and contracts comprise over a quarter of the revenue of these organizations, on average. These two categories combined make up nearly half of the revenue of local Alliance organizations. Events
(Continued page 137)

Advocates celebrating the announcement of upcoming bike lanes on Pennsylvania Avenue in Washington, DC, during the National Bike Summit in 2010. Photo courtesy of Missouri Bicycle and Pedestrian Federation



Number of Residents Per One Member⁽¹⁾ in Statewide Alliance Organizations



Source: Alliance for Biking & Walking 2011 **Notes:** Graph considers only Alliance organizations serving states. As of May 2011, the following states were not served by a dedicated statewide Alliance advocacy organization: Alaska, Kentucky, Louisiana, Montana, Nebraska, North Dakota, and West Virginia. The following states are served by a statewide Alliance advocacy organization that did not provide information on organization revenue: Indiana, Kansas, Ohio, Rhode Island, Texas, and Vermont. Data for Alabama, Idaho, and New Hampshire are from 2009 because no 2010 data were available. Georgia Bikes has free membership, so they do not have any dues-paying members. To see the organization and study area matches, reference Appendix 2, page 200. (1) "Member" is defined as a "dues paying member."

Number of Residents Per One Member⁽¹⁾ in Alliance Organizations Serving Cities



Source: Alliance for Biking & Walking 2011 **Notes:** All data in this chart represent 2010 figures unless otherwise noted. Cities with numbers following indicate there is more than one advocacy organization serving this city. To see the organization and study area matches, reference Appendix 2, page 200. As of May 2011, the following cities were not served by a dedicated local Alliance advocacy organization: Arlington, TX, Baltimore, Colorado Springs, Detroit, El Paso, Fresno, Jacksonville, Mesa, Nashville, Oklahoma City, Phoenix, Raleigh, San Antonio, and Virginia Beach. The following cities are served by a local Alliance advocacy organization that did not provide data: Albuquerque, Las Vegas, Omaha, San Jose, Tucson, and Washington, DC. 2010 membership data unavailable for Cleveland 2, Houston, and Milwaukee so 2009 data were used. (1) "Member" is defined as a "dues paying member."

and sponsorships are another major revenue category for local organizations comprising a third of earned funds on average.

Membership

A strong membership base is often a critical component for successful advocacy organizations. Members provide a donor base, volunteer pool, and can leverage significant political power. Statewide organizations averaged one member per 4,975 people. Maine has the highest rate of members to population with one member for every 195 Maine residents. Organizations serving cities have higher membership rates averaging one member per 1,522 residents. San Francisco has the highest member-

ship rate with one member for every 75 residents. Seattle ranks second with one member for every 154 residents.

New to the 2012 Benchmarking Report, we also collected data on the number of email and mail contacts organizations have. Along with membership, number of contacts indicate the reach an organization potentially has with the community. Both state and local organizations have more mail contacts, on average, than email contacts. Statewide organizations average one mail contact for every 1,297 residents and one email contact for every 1,663 residents. Local organizations average one mail contact for every 268 residents and one email contact for every 299 residents.

Participants of the 2010 Lucinda Means Bicycle Advocacy Day on the steps of Michigan's State Capitol. Many legislators learned about Complete Streets for the first time from bicycle advocates during the annual event. Complete Streets legislation was introduced shortly thereafter and was signed into law with broad bipartisan support in less than six months. Since then, over 50 local Complete Streets ordinances and resolutions have been adopted across the state due to the leadership of bicycle, pedestrian, and disability rights advocates. Photo by John Lindenmayer.



Capacity of Statewide Organizations

State	Year founded	Per capita income 2010	# of staff	Staff per 1 million people	# of members (3)	Residents per one:		
						Member (3)	Mail contact	Email contact
Alabama (2)	2003	\$0.004	0	0.0	100	336,336	13,531	21,115
Arizona	2006	\$0.012	*	*	130	50,737	*	*
Arkansas	2004	\$0.010	0	0.0	37	78,093	4,816	3,853
California	1994	\$0.003	2	0.1	1,200	49,282	14,785	6,160
Colorado	1992	\$0.146	7	1.4	7,205	792	199	586
Connecticut	2005	\$0.046	1.5	0.4	450	7,818	235	1,759
Delaware (2)	2008	\$0.001	*	*	50	17,702	*	*
Florida	1997	\$0.011	2	0.1	2,200	8,426	7,415	4,634
Georgia (2)	2003	\$0.017	1	0.1	0 (4)	Ø	129,332	12,210
Hawaii	1975	\$0.243	5	3.9	500	2,590	2,590	2,590
Idaho (2)	2009	\$0.008	0.5	0.3	42	36,805	16,102	22,732
Illinois	1992	\$0.027	1.8	0.1	1,220	10,582	1,614	5,164
Indiana (2)	1993	\$0.022	*	*	2,000	3,212	*	*
Iowa	2003	\$0.100	3.5	1.2	700	4,297	501	1,203
Maine	1992	\$0.358	4	3.0	6,750	195	131	176
Maryland	2000	\$0.025	1	0.2	60	94,991	1,425	300
Massachusetts	1977	\$0.027	3	0.5	2,827	2,332	758	924
Michigan	1981	\$0.030	2	0.2	2,490	4,004	*	*
Minnesota	2008	\$0.065	3.2	0.6	1,500	1,755	4,389	6,583
Mississippi	1995	\$0.010	0.5	0.2	71	41,577	*	*
Missouri	1993	\$0.008	4	0.7	479	12,500	*	*
Nevada	2009	\$0.0004	0	0.0	10	264,309	105,723	26,431
New Hampshire (2)	2004	\$0.010	2	1.5	89	7,569	2,304	17,661
New Jersey	2009	\$0.005	0	0.0	235	37,054	*	622
New Mexico	2002	\$0.021	1.1	0.5	121	16,609	*	2,512
New York	1995	\$0.004	1.5	0.1	225	86,851	9,771	15,633
North Carolina	2009	\$0.0002	*	*	30	312,696	*	46,904
Oklahoma	1993	\$0.002	0	0.0	130	28,362	*	*
Oregon	1991	\$0.184	16	4.2	6,000	414	109	128
Pennsylvania	2009	\$0.006	1	0.1	700	18,007	21,621	28,199
South Carolina	1998	\$0.015	1	0.2	1,250	3,649	8,293	5,068
South Dakota	2009	\$0.001	*	*	20	40,619	8,124	8,124
Tennessee	2009	\$0.001	*	*	43	146,425	125,925	125,925
Utah	2006	\$0.007	1	0.4	125	22,277	*	11,138
Washington	1987	\$0.105	7	1.1	1,700	3,920	*	*
Wisconsin	1988	\$0.005	10	1.8	3,500	1,616	480	615
Wyoming	1999	\$0.189	1	1.8	1,500	363	605	1,209
Mean/Average (1)	1999	\$0.02	3.0	0.4	7,205	4,975	1,297	1,663
Median	2000	\$0.011	1.8	0.3	450	15,746	4,389	5,068
High	2009	\$0.385	16	4.2	9,250	336,336	129,332	125,925
Low	1975	\$0.0002	0	0.0	0	234	109	128

Source: Alliance for Biking & Walking 2011 Notes: As of May 2011, the following states were not served by a dedicated statewide Alliance advocacy organization: Alaska, Kentucky, Louisiana, Montana, Nebraska, North Dakota, West Virginia, and Wyoming. The following states are served by a statewide Alliance advocacy organization that did not provide data for this report: Indiana, Kansas, Ohio, Rhode Island, Texas, and Vermont. To see the organization and study area matches, reference Appendix 2, page 200. (1) All averages are weighted by population except for year founded, # of members, and # of staff. (2) 2010 data unavailable, 2009 data used. (3) "Member" is defined as a "dues paying member." (4) Georgia has free membership so they do not have any dues-paying members.

Legend:

* = Officials could not access data

Ø = Not applicable

Yellow = High value Light yellow = Low value

CLOSER LOOK

Bicycle Transportation Alliance: Transforming Oregon Communities through Bicycling

(adapted from Bicycle Transportation Alliance)



The Bicycle Transportation Alliance (BTA) is on a mission to transform communities by making bicycling safe, convenient, and accessible. Since 1990, the BTA has grown from a group of activists to an organization with 6,000 individual members, a staff of sixteen, and hundreds of community volunteers.

The BTA has more staff than any state-wide advocacy group and ranks second for per capita income. It may be no coincidence that Oregon has seen greater growth in bicycling than any other state over the last decade.

The BTA's advocacy work is focused on four outcome areas: increase funding, improve policy and safety, build safe and comfortable facilities, and expand the movement.

The BTA engages members directly in advocacy, organizing and activating mem-

bers to speak up in support of bicycle projects, policy, legislation, and funding in their community.

Their advocacy successes include:

- Passing the landmark Portland Bicycle Plan for 2030
- Prevailing in a lawsuit to uphold Oregon's Bicycle Bill
- Protecting the Bicycle Bill from several legislative attacks
- Ensuring that Portland continues to lead with innovative and connected bike facilities

The BTA's work is powered by members and volunteers. Members provide steady financial support and act as the BTA's eyes and ears in the community, helping to monitor advocacy needs. Volunteers are involved in every aspect of the organization, from office help to event coordination to testifying at public hearings.

The BTA also receives funding through individual gifts, corporate sponsorships, contract activities, and government and foundation grants.

"The BTA's work is powered by members and volunteers."

Photos (left to right): A BTA rally for their "Build It" campaign around Portland's Bicycle Plan; The BTA hosts Secretary of Transportation Ray LaHood (middle) and Congressman Earl Blumenauer (far left); A volunteer services a bike at a BTA Service Station, an ongoing outreach and education program of the BTA. Photos courtesy of the Bicycle Transportation Alliance, Katy Cannatelli, and Natalie Baker.



Capacity of Organizations Serving Cities

City	Year founded	Per capita income 2010	# of staff	Staff per 1 million people	# of members (3)	Residents served per one:		
						Member (3)	Mail contact	Email contact
Atlanta	1991	\$0.15	1.5	1.5	550	1,818	1,429	200
Austin	1995	\$0.02	1	0.7	240	6,250	6,250	714
Boston 1	2005	\$0.11	1	1.3	*	*	2,000	182
Boston 2	1990	\$0.07	3	0.5	238	27,704	1,379	2,484
Boston 3	2010	\$0.02	0.2	0.3	200	3,400	378	272
Charlotte	1998	\$0.03	0.5	1.0	15	33,333	10,000	208
Chicago	1985	\$0.36	40	4.7	*	*	113	213
Chicago 2	2004	\$0.0002	*	*	*	*	ø	6,076
Cleveland 1	2006	\$0.04	1.5	1.5	*	*	*	200
Cleveland 2 (2)	2003	\$0.01	0	0.0	45	22,222	*	*
Columbus 1	1991	\$0.10	3	1.2	200	12,500	2,500	1,000
Columbus 2	2008	\$0.00	0	0.0	0	ø	*	1,786
Dallas	2008	\$0.002	0	0.0	40	50,779	*	*
Denver	2001	\$0.04	2	0.9	140	16,786	4,700	1,567
Fort Worth	2008	\$0.002	0	0.0	40	50,779	*	*
Honolulu	2009	\$0.24	5	3.9	500	2,590	2,590	2,590
Houston (2)	2004	\$0.001	0	0.0	46	49,150	*	*
Indianapolis 1	1985	\$0.16	1	1.2	*	*	*	*
Indianapolis 2	2010	\$0.01	2.5	3.0	120	6,914	4149	3319
Kansas City, MO	1975	\$0.01	0	0.0	300	6,667	6,667	2,353
Long Beach	1998	\$0.06	8.5	0.9	900	11,111	*	1,333
Los Angeles	1998	\$0.06	8.5	0.9	900	11,111	*	1,333
Louisville	2005	\$0.07	0.5	0.9	*	*	246	233
Memphis	2005	\$0.06	1	1.0	*	*	16,667	1,250
Miami	2007	\$0.07	0.5	1.2	104	4,165	2,166	866
Milwaukee	1988	\$0.01	10	1.8	3,000(2)	1,885(2)	480	615
Minneapolis 1	1990	\$0.56	4	6.0	*	*	149	134
Minneapolis 2	1992	\$0.60	3	7.8	821	469	82	107
New Orleans	2003	\$0.07	1	3.3	91	3,297	536	197
New York	1973	\$0.27	20	2.4	8,500	1,000	850	170
Oakland	2007	\$0.23	0.6	1.5	422	926	269	242
Philadelphia	1972	\$0.17	12	2.0	1,400	4,286	857	667
Portland, OR	1991	\$0.18	16	4.2	6,000	414	109	128
Sacramento 1	1991	\$0.08	1.6	1.2	900	1,444	1,000	1,857
Sacramento 2	1998	\$0.11	4	2.9	*	*	7,000	4,667
San Diego	1993	\$0.08	2	1.5	1,100	1,187	*	*
San Francisco	1971	\$1.61	11	13.6	10,747	75	16	14
Seattle 1	1970	\$1.60	25.5	12.8	13,000	154	36	31
Seattle 2	2001	\$0.12	2	1.0	61	32,787	33,333	33,333
Seattle 3	2007	\$0.02	1	0.5	*	*	*	571
Tulsa	2008	\$0.16	0	0.0	*	*	973	1,947
Mean/Average (1)	1996	\$0.15	4.9	2.2	1,796	1,522	268	299
Median	1998	\$0.07	1.6	1.2	270	4,225	987	667
High	2010	\$1.61	40	33.3	13,000	50,799	33,333	33,333
Low	1970	\$0.0002	0	0.00	0	75	16	14

Legend:

* = Officials could not access data

ø = Not applicable

Yellow = High value

Light yellow = Low value

Source: Alliance for Biking & Walking 2011 **Notes:** All data in this chart represent 2010 figures unless otherwise noted. As of May 2011, the following cities were not served by a dedicated local Alliance advocacy organization: Arlington, Baltimore, Colorado Springs, Detroit, El Paso, Fresno, Jacksonville, Mesa, Nashville, Oklahoma City, Phoenix, Raleigh, San Antonio, and Virginia Beach. The following cities are served by a local Alliance advocacy organization that did not provide data: Albuquerque, Las Vegas, Omaha, San Jose, Tucson, Washington, DC. Cities with numbers following indicate there is more than one advocacy organization serving this city. To see the organization and study area matches, reference Appendix 2, page 200. (1) All averages are weighted by population except for year founded, # of members, and # of staff. (2) 2010 data unavailable, 2009 data used. (3) "Member" is defined as a "dues paying member."

CLOSER LOOK



Cascade Bicycle Club: Seattle's Voice for Biking

Seattle-based Cascade Bicycle Club (CBC), founded in 1970, is one of the nation's oldest and largest bicycle clubs. The CBC has grown to be an advocacy powerhouse with 13,000 members, one for every 154 Seattle residents. The organization operating budget is \$1.60 per capita making it one of the best-funded local advocacy organizations in the United States.

We asked Chuck Ayers, Executive Director of CBC, to help explain why his organization is thriving.

Alliance: *What factors do you feel have most contributed to CBC's success?*

Chuck: First, unlike many bicycle-focused organizations or riding clubs, Cascade is a multifaceted, multi-programmatic organization. Our programs include daily rides (over 180 Cascade "certified" ride leaders leading over 1,200 free-to-the-public rides annually); professionally produced cycling-related fundraising events; youth and adult education including classes and week-long summer camps for kids; business-focused commute programs including a bike to work day event drawing over 20,000 participants and a month-long commute challenge with over 11,000 participants; and advocacy (policy, planning, government affairs). Our programmatic bandwidth gives cyclists multiple entry points into the organization—and into our database! This then gives us the opportunity to reach, educate, and activate tens of thousands of cyclists and expand our brand recognition.

Second, we endorse political candidates. Political strength isn't just about



Cascade Bicycle Club's annual Chilly Hilly ride attracts thousands of bicyclists to Bainbridge Island, WA (just outside of Seattle). Photo courtesy of Cascade Bicycle Club.

"Our programmatic bandwidth gives cyclists multiple entry points into the organization—and into our database!"

who's on your side and who you can lobby. It's about who you can help put in office. Our membership and constituents can turn an election and thus our endorsement is highly valued at the local, regional, and state level. From there, it's about accountability.

Third, we do a lot of policy and bike skills education which creates wide constituency support and a platform on

which to stand when anti-bikers complain about cyclists (their riding, their “not paying their fair share,” etc.).

Finally, we hire great people and let them do their jobs in a fun and supportive environment.

Alliance: *Why is CBC a leader among advocacy organizations nationwide?*

Chuck: We have professional staff who are knowledgeable, passionate, and capable of creating and executing a variety of campaigns and strategies. We have built strong relationships with elected officials, governmental agencies, and community leaders. We have built a very large, knowledgeable, and activist constituency. And we engage in the electoral process through political endorsements, campaign contributions, and independent expenditures.

We are organized for political action. Cascade Bicycle Club is an IRS designated 501(c)(4) organization and we have an affiliated Political Action Committee (BikePAC). (The Cascade Bicycle Club Education Foundation is an IRS designated 501(c)(3) organization.)

Alliance: *How does CBC use revenue and membership to make the Seattle area a better place to bike?*

Chuck: We are not just in Seattle but rather work regionally and at the state legislature. We invest in effective broad-based programming (daily rides, advocacy, education, planning, and commuting); we keep members informed and educated and they in return trust us and a large proportion are willing to engage in planning and political processes; we engage the business community; and we have fun.

After three years of working with the Washington State Legislature, CBC's efforts paid off on May 16, 2011 when Gov. Chris Gregoire signed the Vulnerable User Bill into law. The law goes into effect on July 1, 2012 and increases penalties for negligent driving that injures or kills a vulnerable roadway users like pedestrians and bicyclists. Prior to the bill's enactment, a minor traffic ticket was the greatest penalty, even in the case of a fatality. Photo courtesy of Washington State Senate Photography.





Park on the Atlanta BeltLine. Photo by Andrew Dannenberg.

7: INFLUENCING BIKING AND WALKING

We are constantly learning what influences bicycling and walking. Studies show that a number of factors contribute to the choice to bicycle or walk (Heinen et al., 2010, Krizek and Forsyth 2009, and Pucher, Dill, Handy 2010). These include environmental influences such as climate, density, mix of uses, and infrastructure; socioeconomic and demographic factors such as age, gender, and car ownership; and other factors including education and existing levels of bicycling and walking.

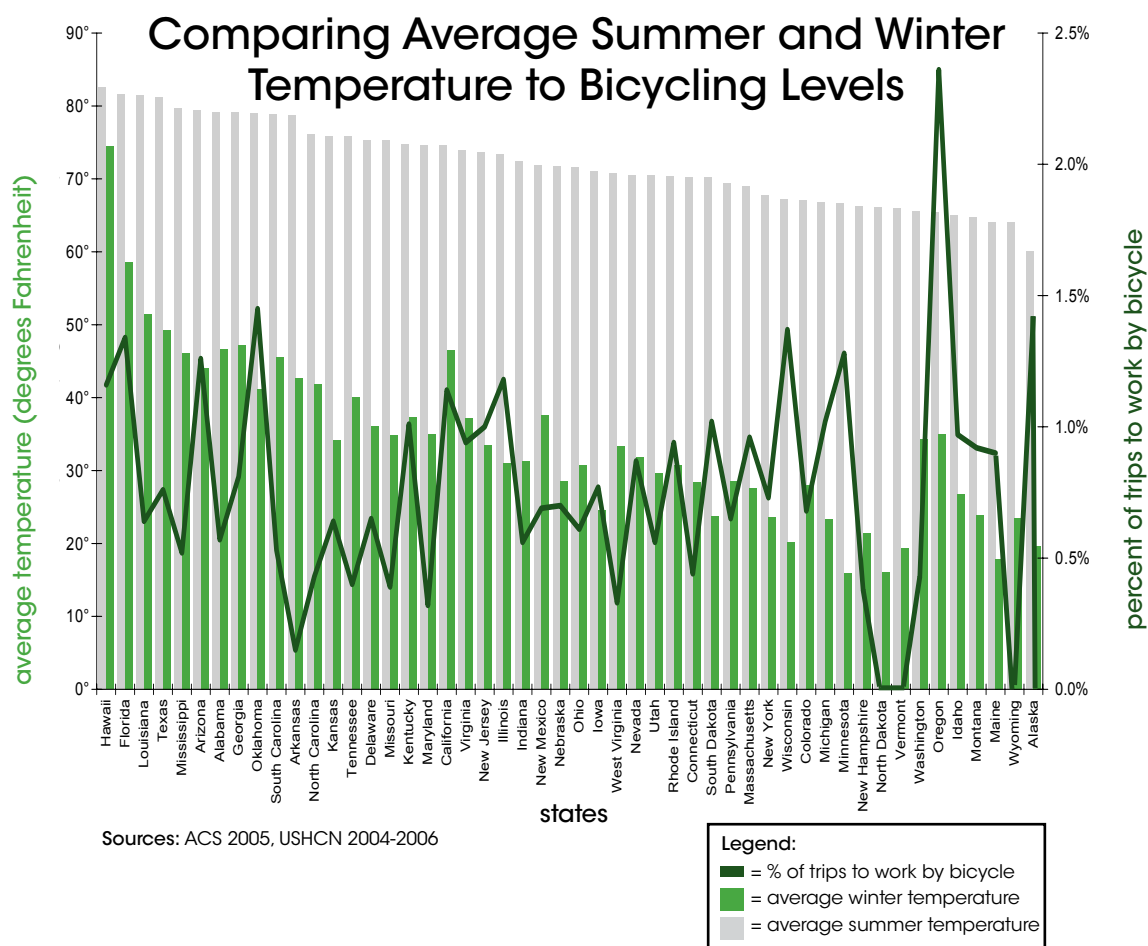
This chapter looks at a few potential relationships using data from previous chapters and additional data from the Census and American Community Survey. Also included is a closer look at Oregon, the state with the greatest

recent growth in bicycling, and Seville, Spain, a country that has experienced a recent boom in bicycling levels. These two examples demonstrate how policy decisions and investment can play a vital role in determining how many people bicycle and walk.

Environmental Influences

Climate

Does climate influence the choice to bicycle? The Alliance compared average summer and winter temperatures to bicycle share of work trips in 50 states and did not find any compelling evidence that climate is a major influence. Montana and Alaska, for example, are



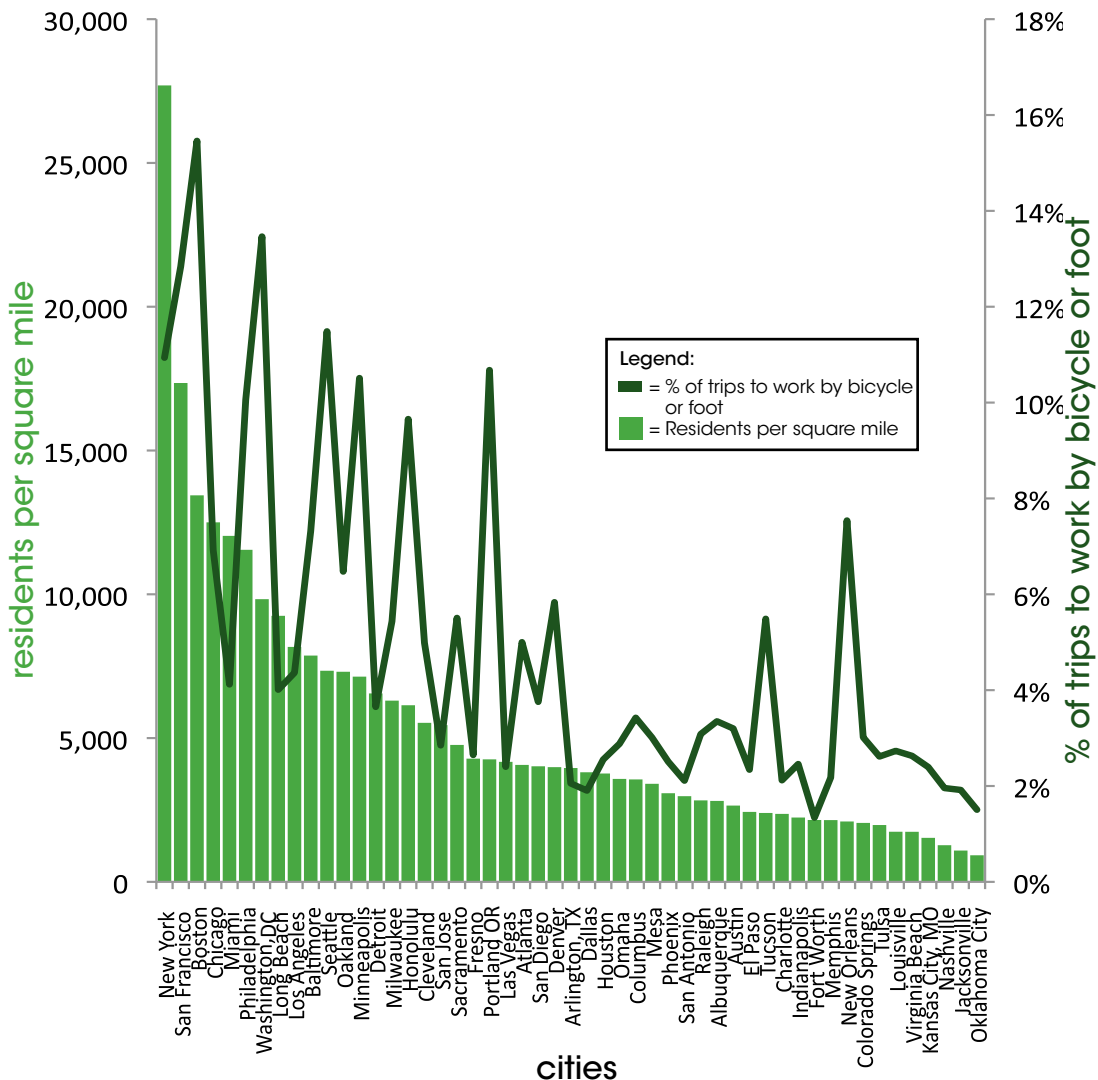
among states with the coldest temperatures, yet are also among the states with the highest levels of bicycling. The lack of statistically significant evidence of climate's impact on bicycling levels has been noted in other studies (Buehler and Pucher 2011, Heinen et al., 2010, Krizek and Forsyth 2009, Pucher and Buehler, 2006, Pucher et al., 2011) pointing out much higher rates of bicycling in countries such as Canada, with lower average year-round temperatures than in the United States. It seems likely that excessive cold, heat, and rainfall do indeed deter bicycling to some unknown extent, especially among less dedicated bicyclists. According to a poll by the San Francisco Bicycle Coalition for their Report Card on Bicycling, 11% of respondents said that weather kept them from bicycling more than they

do (down from 15% of respondents in 2006, two years earlier). Concerns about bicycle theft, safety, and lack of carrying capacity were the other top reasons cited for not bicycling more.

Density

To examine the role of density in the choice to bicycle or walk in the United States, the Benchmarking Project team compared residential density (persons/square mile) to the combined bicycling and walking to work mode share in major cities. Data indicate that denser cities have higher levels of bicycling and walking on average than less dense cities. Four of the five cities with the highest combined levels of bicycling and walking are also among the top seven densest cities. This finding is in line with other studies (Heinen et al.,

Residential Density and Bicycling and Walking Levels in Major U.S. Cities

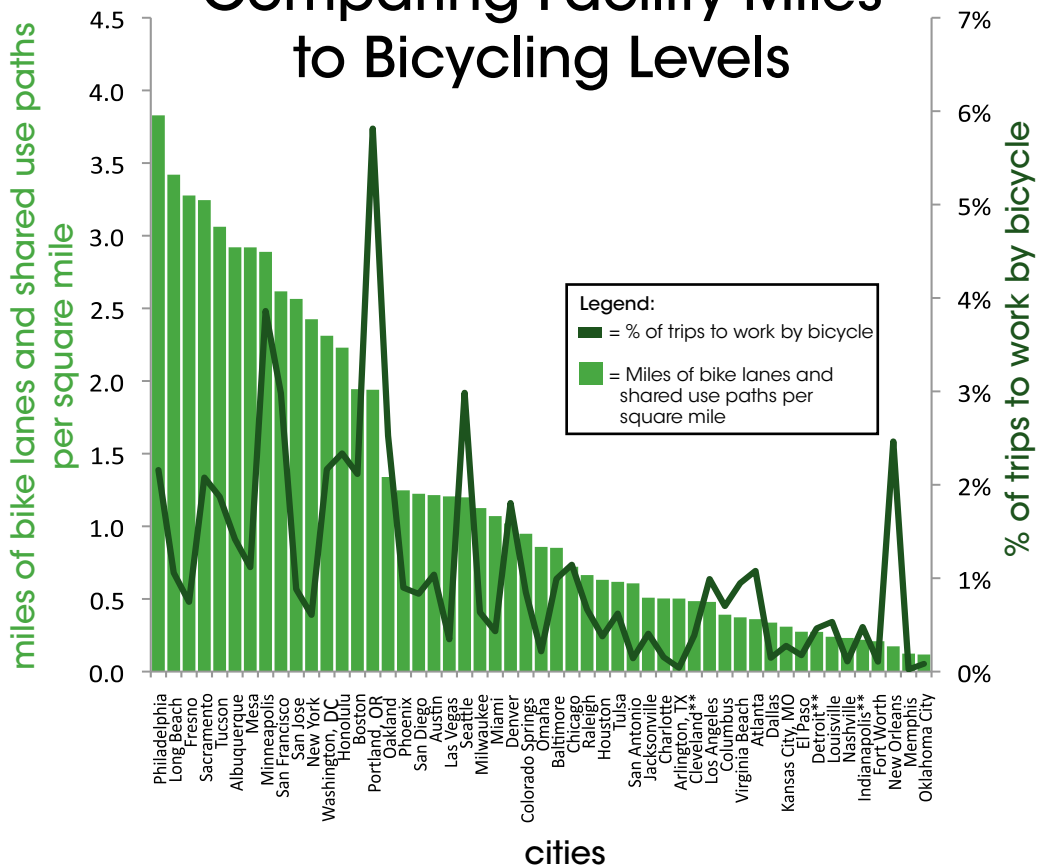


Source: ACS 2009 Note: $r = 0.69$.

Cities with the most residents per square mile have higher levels of bicycling and walking, on average, than less dense cities. Boston, Washington, DC, San Francisco, and New York, four of the five cities with the highest combined rates of bicycling and walking, are also among the top seven densest cities. The least dense cities, including Jacksonville and Nashville, are among the cities with the lowest levels of bicycling and walking.

**Denser cities
have higher rates
of bicycling and
walking.**

Comparing Facility Miles to Bicycling Levels



Sources: ACS 2009, City surveys 2010/2011 Note: $r = 0.51$; Cities with ** = miles of bike lanes and paths per sq mile is from 2010.

2010, Krizek and Forsyth 2009, Moudon et al., 2005, Parkin et al., 2008, Pucher and Buehler, 2006, Pucher et al., 2011, Reynolds et al., 2009, Rietveld and Daniel 2004, Vandenbulck et al., 2011, Vernez-Moudon et al., 2005) that suggest a correlation between density and bicycling and walking. Dense communities have shorter trip distances, which can thus be more easily covered by walking or bicycling.

Walking and Bicycle Facilities

The extent and quality of bicycle and pedestrian facilities almost certainly affect levels of bicycling and walking (Buehler and Pucher 2011, Dill and Carr 2003, Heinen et al., 2010, Hunt

and Abraham 2007, J Moudon et al., 2005, Parkin et al., 2008, Pucher et al., 2010, Rietveld and Daniel 2004, Vandenbulcke et al., 2011), but the available data are limited and do not indicate an obvious relationship. Because there is no standard reporting requirement for government agencies to track bicycle and pedestrian facilities, many do not have accurate records. Also, the quality and accessibility of facilities are difficult to measure and may vary greatly from place to place. For example, a 12-foot-wide multi-use path on a major city bridge may be much more important for increasing bicycling and walking by providing a network link than a 4-foot-wide path through a small neighborhood. U.S. bicycle advo-

cates commonly look to places like the Netherlands where cities have invested heavily in an infrastructure network for bicycling. These investments (including bike lanes, separated paths, and specialized signals and traffic signs for bicyclists) may contribute to a bicycling mode share that reaches between 30% to 50% in many Dutch cities (Pucher and Buehler, 2007 and 2008). This report compared miles of bicycle facilities per square mile to levels of bicycling in cities. Results suggest there may be a relationship between facilities and mode share. Although it is not true in every case, the general trend is that cities with higher levels of bicycling have more bicycle facilities per square mile than cities with lower bicycling levels.

Socioeconomics and Demographics

Income

Income could play a role in the decision to bicycle or walk (Buehler and Pucher 2011, Shafizadeh and Niemeier 1997, Vandenbulcke et al., 2011). As graphs in Chapter 3 show, bicycling levels are fairly even among various income groups although the majority of pedestrians earn less than \$35,000 a year. Among states, 46% of people who walk to work earn less than \$15,000 a year. Nearly 3/4 of people who walk to work nationwide earn less than \$35,000 a year. Although income may play a role in the decision to walk to work for some people, among cities with higher levels of walking, such as New York, income levels are more evenly distributed among pedestrians. This suggests that income alone is not an important factor

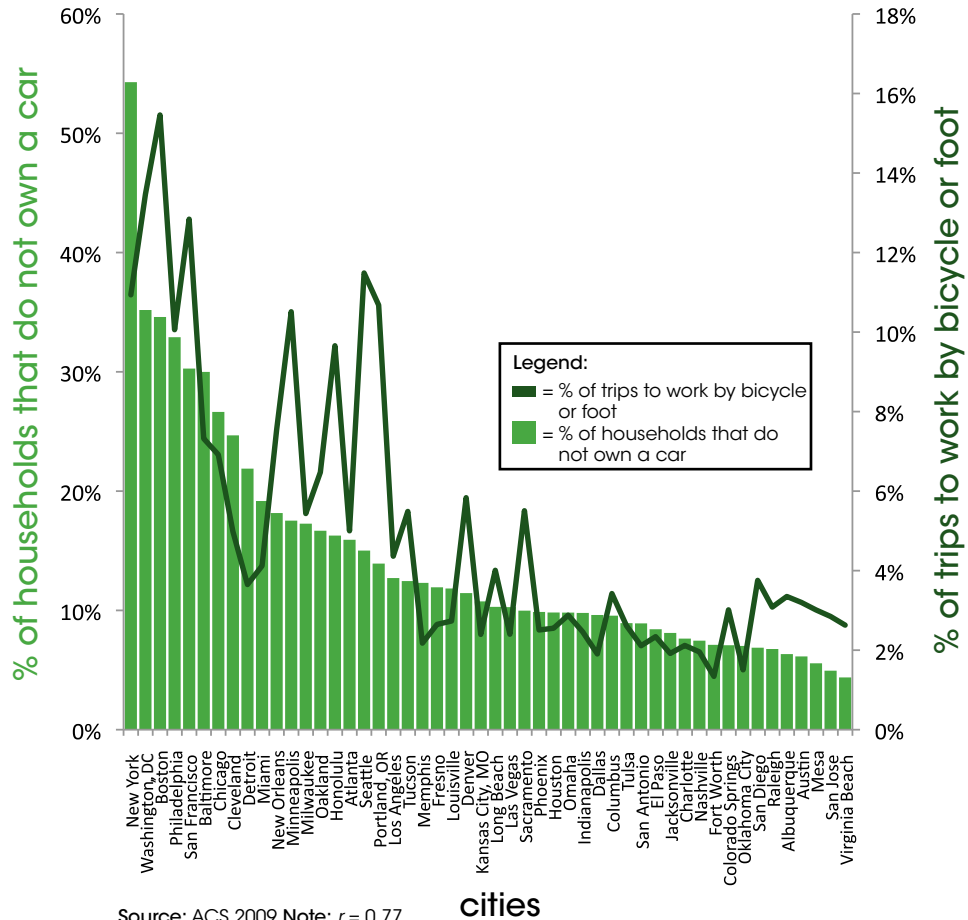


in the decision to walk in dense transit-oriented cities.

Car Ownership

Owning a car is definitely related to levels of walking and bicycling (Vandenbulcke et al., 2011). According to the 2009 ACS, cities with the highest levels of bicycling and walking have the lowest car ownership rates. Although the statistical relationship is strong ($r = 0.77$), the causation probably runs in both directions. Those who walk or bicycle a lot are less likely to need or want a car. And those who do not own a car are more likely to need to walk or bicycle for some trips. In any case, it is clear that high levels of car ownership are strongly related to low levels of walking and bicycling.

Comparing Car Ownership to Bicycling and Walking Levels



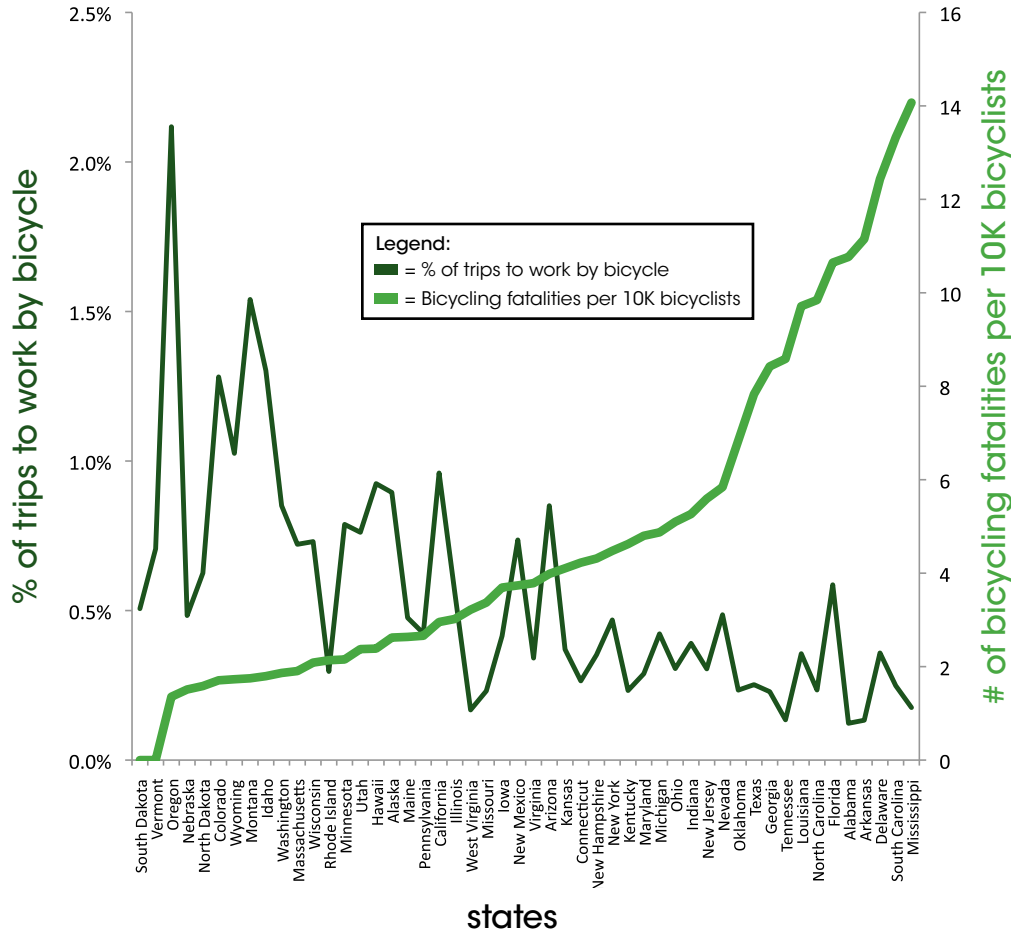
Levels of Bicycling, Walking, and Safety

To see how levels of bicycling and walking affect safety, the Benchmarking Project team compared fatality data reported by cities to ACS 2009 bicycle and pedestrian mode share (trips to work). Results were consistent with previous research (Elvik 2009, Jacobsen 2003, Pucher et al., 2011, Vandenbulcke et al., 2009) indicating a negative correlation between levels of bicycling and walking and fatality rates. Cities with the highest levels of bicycling generally have

lower bicycle fatality rates. Cities with the highest rates of pedestrian fatalities are also among those with the lowest levels of walking. A possible explanation could be that in places where more bicyclists and pedestrians are present, motorists are more used to sharing the roadways with bicyclists and are more aware of pedestrians at crossings. Environmental factors that contribute to increased bicycling and walking (such as signed routes, bike lanes, and side-walks) may also contribute to increased safety.

(Continued page 152)

Relationship between Bicycle Fatalities and Bicycling Levels

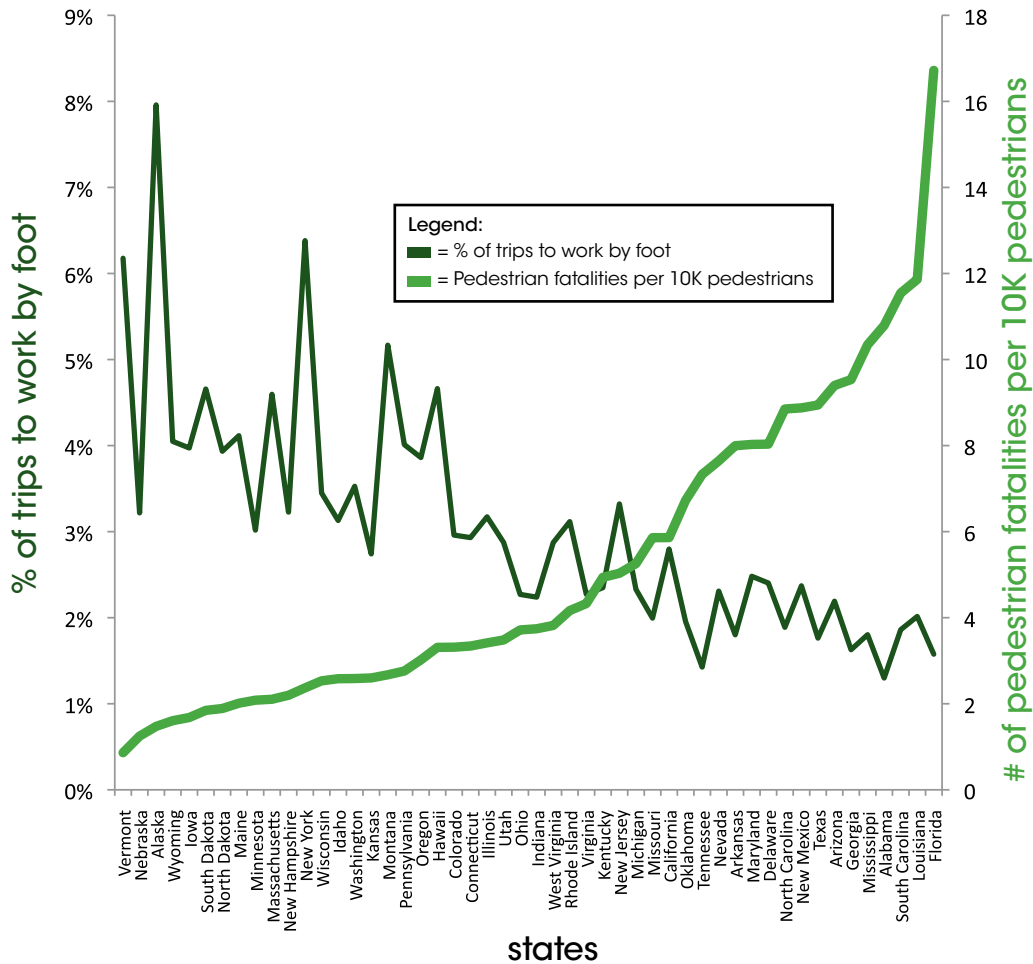


Sources: ACS 2009, FARS 2007-2009 Notes: Bicyclist fatality rate was calculated by dividing the number of annual bicycle fatalities by population (weighted, or multiplied, by share of the population biking to work). Because of the approximate nature of the exposure data and great fluctuations in fatality data from year to year, this rate should be seen as a rough estimate and not the literal number of fatalities per 10,000 bicyclists. $r = -0.54$.

Many studies show that bicycling safety increases greatly as bicycling levels rise (Elvik 2009, Jacobsen 2003, Pucher et al., 2011). For this illustration, bike to work mode share from ACS 2009 was compared to FARS bicycle fatality data (using a 3-year average number of fatalities). To figure the fatality rate, the Alliance divided the number of fatalities by population (weighted, or multiplied, by share of the population biking to work) (to more accurately determine exposure levels). The result is a negative correlation ($r = -0.52$) that suggests greater bicycling levels may mean increased safety for bicyclists.

**Bicycle
safety
increases
with more
bicycling.**

Relationship between Pedestrian Fatalities and Walking Levels



Sources: ACS 2009, FARS 2007-2009 **Notes:** Pedestrian fatality rate was calculated by dividing the number of annual pedestrian fatalities by population (weighted, or multiplied, by share of the population walking to work) (to adjust for exposure). Because of the approximate nature of the exposure data and great fluctuations in fatality data from year to year, this rate should be seen as a rough estimate and not the literal number of fatalities per 10,000 pedestrians. $r = -0.69$.

Higher levels of walking contribute to pedestrian safety.

States with higher levels of walking have lower rates of pedestrian fatalities. For this illustration, pedestrian worker mode share from the ACS was compared to FARS data on pedestrian fatalities. To figure fatality rate, the Alliance divided the number of fatalities by population (weighted, or multiplied, by share of the population walking to work) (to more accurately determine exposure levels). The result is a negative correlation ($r = -0.69$) that suggests greater walking levels may mean increased safety for pedestrians.



Advocacy and Education

Advocacy's Impact

Chapter 6 discussed the challenges presented by attempting to quantify advocacy. Every bicycle and pedestrian advocacy organization is different in structure and operations. A number of the organizations surveyed have been around for a decade or more, while others are only 1 to 2 years old. Some organizations are volunteer-run, while others have teams of full-time staff.

This report compared per capita income (organization revenue/city population) and staffing levels of organizations to levels of bicycling and walking. Results indicate a positive correlation between levels of bicycling and walking to work and the standardized income ($r = 0.62$) and staffing levels ($r = 0.59$) of Alliance organizations. Although one cannot assume that advocacy capacity and bicycling and walking levels are causally related, comparing the two at least suggests that the presence of a strong advocacy organization can be an indicator of a city's bicycling and walking levels. Causation could go in either direction. Cities with higher bicycling and walking rates are likely to have more people supportive of advocacy, and cities with strong advocacy organizations are likely to experience growth in bicycling and walking.

Education's Impact

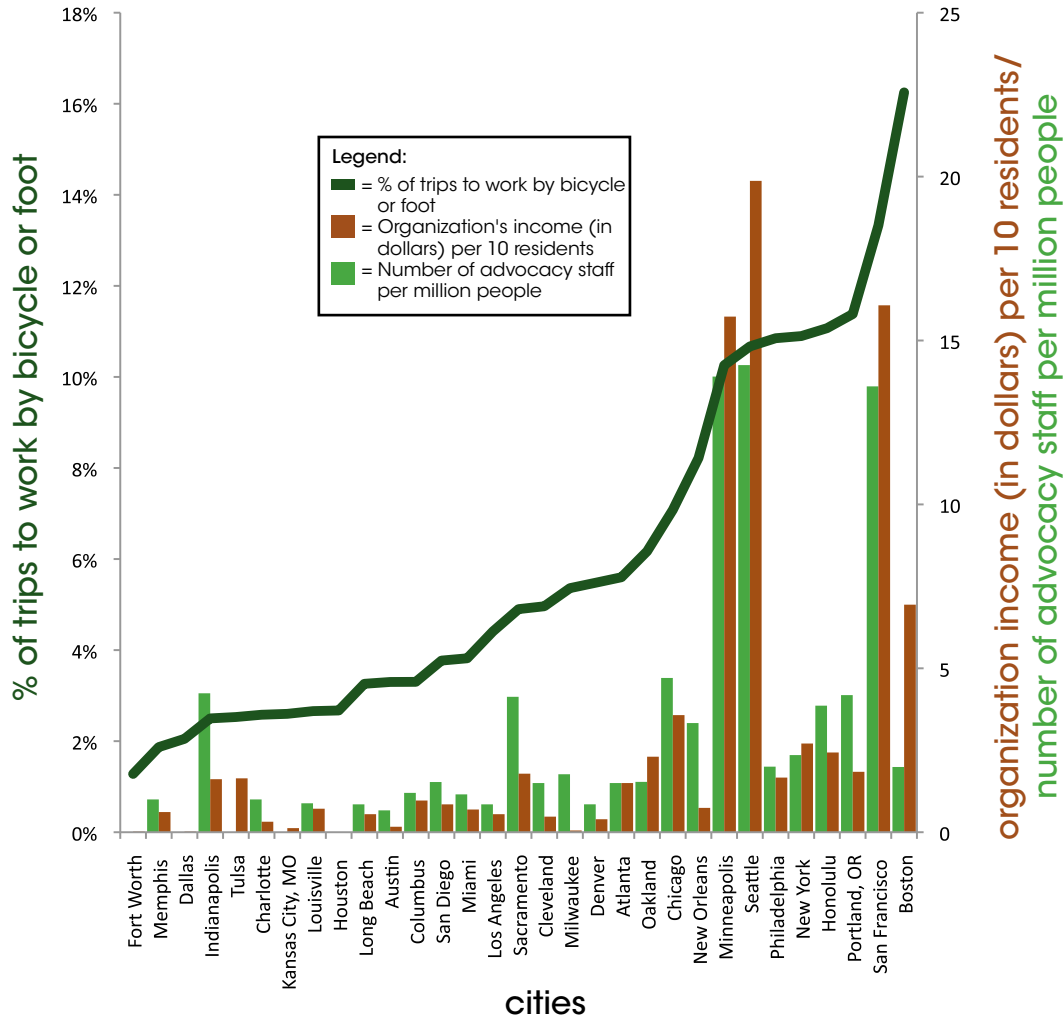
Another potential factor contributing to bicycling, walking, and safety is education. As demonstrated in Chapter 5, many advocacy organizations and government agencies sponsor education and encouragement efforts which may influence mode share and safety. Although some baseline data were

(Continued page 156)



Girl practices bicycling skills at the San Francisco Bicycle Coalition's annual Family Day in Golden Gate Park. Photo by Kate McCarthy

Relationship between Advocacy Capacity and Mode Share



Sources: ACS 2009, Alliance for Biking & Walking 2011 **Notes:** $r = 0.62$ (organization income per 10 residents/bike + walk levels) $r = 0.59$ (organization staffing per million residents/bike + walk levels).

The Alliance compared the combined levels of bicycling and walking to work from the ACS 2009 to standardized income and staffing levels of Alliance organizations serving these cities. Positive correlations ($r = 0.62$ and $r = 0.59$) suggest that a relationship may exist between advocacy capacity and levels of bicycling and walking.

**Advocacy capacity
may be linked to
higher levels of
bicycling and
walking.**

CLOSER LOOK

Oregon's Rising Levels of Bicycling

by Susan Peithman, Bicycle Transportation Alliance



Oregon is a safe and wonderful place to ride a bicycle. In terms of culture, infrastructure, and politics, Oregon is welcoming and encouraging to bicyclists. It is no wonder that bicycling's popularity has grown faster here than anywhere in the United States. Between 2000 and 2009, the share of commuters who bicycle to work increased from 1.07% to 2.34%, a larger jump than in any other state. Oregon also saw a 193% increase in bicycle commuters between 1990 and 2009—the greatest increase among states.

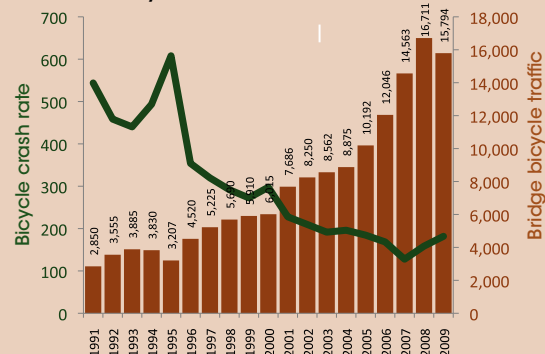
Of all 50 states, Oregon has the greatest percentage of commuters who bike to work (2.1%). Of these bicycle commuters, 33% are women, significantly higher than the national average of 26% and second only to Montana where 34% of bicycle commuters are women.

The visibility of bicycling encourages even more people to ride and makes roads safer, as drivers and all road users are aware of bicyclists. In Portland, where the amount of cyclists has doubled over the past decade, the number

of crashes involving a person on a bike has remained relatively constant. This trend indicates the roads are becoming safer as more people ride bikes (see chart below).

Oregon's rapidly increasing bicycle use is largely a product of the state's already prominent bicycle culture, which encompasses everything from casual riders to racing teams and "ZooBombers." Throughout the state, and especially in the city of Portland, bicycles

Portland Bicycle Bridge Traffic Versus Bicycle Crashes 1991-2010



Source: Portland Bureau of Transportation
2009 Bicycle Count

Legend:

- = Bicycle crash rate (crashes per 10K bicycle bridge riders)
- = Bridge bicycle traffic (number of bicyclists on count day)

Eugene, OR Kidical Mass is a testament to a diverse and thriving bicycle culture in the state.
Photo by Shane MacRhodes





Corvallis, OR Bike Path Ride. Photo by Dan Crouch

intermingling with cars and pedestrians are commonplace.

Oregon has been inviting to bicyclists as early as the 1970s. Because of a 1973 law establishing urban growth boundaries on cities, destinations in Oregon's urban areas are built close to each other in a well-organized grid layout. This planning encourages the 2- to 3-mile trips convenient for bicycle travel. Additionally, Oregon's "Bicycle Bill" of 1971 mandates bicycle accommodations in all transportation facilities and ensures that at least 1% of transportation funding is devoted to bicycling infrastructure.

Oregonians find it easier to pick up bicycling than people in most other

states because of the relative abundance of bicycle friendly infrastructure and government policy. Portland alone has over 325 miles of bike paths, and bike racks are available at any major destination. Even Portland's stoplight timings are set to slow cars making streets safer for bicyclists. This dedication to promoting bicycling was recently further bolstered by the Jobs and Transportation Act of 2009, which supports green and active transportation. Consequently, people considering making the switch from cars to bicycles find it convenient, safe, and enjoyable.

Progressive bicycle legislation is possible in Oregon because bicyclists are represented by well-staffed and well-funded advocacy groups. The statewide advocacy organization, the Bicycle Transportation Alliance (BTA), has the highest number of staff per capita served (4.2 per 1 million people) and has 6,000 members, ranking third among statewide organizations for membership. The BTA is influential in urban politics and is responsible for many of Oregon's bicycling improvements. Its Safe Routes to Schools program, for example, operates at over 70 schools.

Though Oregon is already America's leader in bicycle culture, the future of the state's bicycle policy is ever progressive and ambitious. Portland, working closely with the BTA, finalized a 20-year, \$613 million plan for improvements to its bicycling infrastructure. The plan, the nation's most ambitious, calls for 368 miles of on-road bikeways, 78 miles of bike trails, and 256 miles of bicycle boulevards. Oregon is committed to growing its population of bicyclists in the years to come.